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Distributing certificate bundles according to fault domains

Abstract

Operations of a certificate bundle distribution service may include: detecting a trigger condition to distribute a certificate bundle that includes a set of certificate authority certificates; determining, for each of a plurality of network entities associated with a computer network, a fault domain representing at least one single point of failure; partitioning the plurality of network entities into a plurality of certificate distribution groups, based on a set of partitioning criteria that includes a fault domain of each particular network entity, in which each particular certificate distribution group includes a particular subset of network entities, and the particular subset of network entities are associated with a particular fault domain; selecting a particular certificate distribution group, of the plurality of certificate distribution groups, for distribution of the certificate bundle; and transmitting the certificate bundle to the particular subset of network entities in the particular certificate distribution group.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5699431	12/1996	Van Oorschot et al.	N/A	N/A
7272714	12/2006	Nagaratnam et al.	N/A	N/A
7644270	12/2009	Cherukumudi et al.	N/A	N/A
8452958	12/2012	Sun et al.	N/A	N/A
9172543	12/2014	Wnuk	N/A	N/A
9197630	12/2014	Sharif et al.	N/A	N/A
9231933	12/2015	Shenoy et al.	N/A	N/A
9485101	12/2015	Bowen	N/A	N/A
9660978	12/2016	Truskovsky et al.	N/A	N/A
9680813	12/2016	Sade et al.	N/A	N/A
9794249	12/2016	Truskovsky et al.	N/A	N/A
9882727	12/2017	Veladanda et al.	N/A	N/A
10021084	12/2017	Matthews et al.	N/A	N/A
10212147	12/2018	Buendgen et al.	N/A	N/A
10652030	12/2019	Levy et al.	N/A	N/A
10764263	12/2019	Rossi	N/A	N/A
10771261	12/2019	Lazar et al.	N/A	N/A
10812276	12/2019	Bojjireddy et al.	N/A	N/A
10848323	12/2019	Barr, III et al.	N/A	N/A
11153103	12/2020	Fynaardt et al.	N/A	N/A
11196570	12/2020	Borne-Pons et al.	N/A	N/A
11310059	12/2021	Leibmann et al.	N/A	N/A
11362843	12/2021	Jiang et al.	N/A	N/A
11368314	12/2021	Ray et al.	N/A	N/A
11388594	12/2021	Uy et al.	N/A	N/A
11438325	12/2021	Begun et al.	N/A	N/A
11627123	12/2022	Stayskal et al.	N/A	N/A
11706038	12/2022	Thakore et al.	N/A	N/A
11888997	12/2023	Bowen et al.	N/A	N/A
12088738	12/2023	Rosenthol et al.	N/A	N/A
2002/0007346	12/2001	Qiu et al.	N/A	N/A
2002/0174066	12/2001	Kleckner et al.	N/A	N/A
2003/0037234	12/2002	Fu et al.	N/A	N/A
2006/0047965	12/2005	Thayer	N/A	N/A

2006/0101510	12/2005	Kadyk et al.	N/A	N/A
2007/0005956	12/2006	Zilinskas et al.	N/A	N/A
2007/0147619	12/2006	Bellows et al.	N/A	N/A
2010/0030897	12/2009	Stradling	N/A	N/A
2010/0325429	12/2009	Saha et al.	N/A	N/A
2011/0113239	12/2010	Fu et al.	N/A	N/A
2012/0036220	12/2011	Dare et al.	N/A	N/A
2012/0246466	12/2011	Salvarani et al.	N/A	N/A
2014/0298419	12/2013	Boubez et al.	N/A	N/A
2015/0135299	12/2014	Liang et al.	N/A	N/A
2017/0039373	12/2016	Sasin et al.	N/A	N/A
2017/0126667	12/2016	Bishop et al.	N/A	N/A
2017/0171191	12/2016	Cignetti et al.	N/A	N/A
2017/0317837	12/2016	Alrawais et al.	N/A	N/A
2017/0338967	12/2016	Lewison et al.	N/A	N/A
2018/0083966	12/2017	Zhou et al.	N/A	N/A
2018/0102904	12/2017	Lin et al.	N/A	N/A
2018/0287804	12/2017	Geisbush	N/A	N/A
2019/0149342	12/2018	Fynaardt et al.	N/A	N/A
2019/0165950	12/2018	Ibrahim	N/A	H04L 9/3268
2019/0347406	12/2018	Lev-Ran	N/A	N/A
2019/0349402	12/2018	Shukla et al.	N/A	N/A
2019/0363895	12/2018	Barr et al.	N/A	N/A
2020/0021575	12/2019	Rezvani et al.	N/A	N/A
2020/0092095	12/2019	Yang et al.	N/A	N/A
2020/0150972	12/2019	Ketkar et al.	N/A	N/A
2020/0274718	12/2019	Hwang et al.	N/A	N/A
2020/0274862	12/2019	Varvarezis et al.	N/A	N/A
2021/0034767	12/2020	Free et al.	N/A	N/A
2021/0126801	12/2020	Nix	N/A	G06F 7/588
2021/0152547	12/2020	Barhudarian et al.	N/A	N/A
2021/0211307	12/2020	Statia et al.	N/A	N/A
2021/0218723	12/2020	Lekov et al.	N/A	N/A
2021/0392002	12/2020	Gray et al.	N/A	N/A
2021/0409403	12/2020	Lewin et al.	N/A	N/A
2021/0409409	12/2020	Palanisamy	N/A	N/A
2022/0038894	12/2021	Yoon et al.	N/A	N/A
2022/0123951	12/2021	Lutz et al.	N/A	N/A
2022/0150238	12/2021	Bhalerao	N/A	N/A
2022/0239503	12/2021	Mallikarjuna et al.	N/A	N/A
2022/0393886	12/2021	Williams et al.	N/A	N/A
2023/0032867	12/2022	Peddada et al.	N/A	N/A
2023/0109231	12/2022	Adogla et al.	N/A	N/A
2023/0208655	12/2022	Statia et al.	N/A	N/A
2023/0237155	12/2022	Jacquín et al.	N/A	N/A
2023/0291577	12/2022	Thai et al.	N/A	N/A
2023/0401307	12/2022	Pop	N/A	G06F 8/65
2023/0412397	12/2022	Gollent et al.	N/A	N/A
2024/0015508	12/2023	Yoon et al.	N/A	N/A

2024/0020373	12/2023	Ivanov et al.	N/A	N/A
2024/0031146	12/2023	Marosi-Bauer et al.	N/A	N/A
2024/0106886	12/2023	Roy et al.	N/A	N/A
2024/0121603	12/2023	Yoon et al.	N/A	N/A
2024/0146543	12/2023	Sahoo et al.	N/A	N/A
2024/0333640	12/2023	Shevade et al.	N/A	N/A
2024/0356763	12/2023	Goldberg et al.	N/A	N/A
2024/0388510	12/2023	Madtha et al.	N/A	N/A
2025/0030561	12/2024	Long et al.	N/A	N/A
2025/0088373	12/2024	Uzun et al.	N/A	N/A
2025/0133401	12/2024	Lee et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112019477	12/2019	CN	N/A
114884963	12/2021	CN	N/A
1251670	12/2001	EP	N/A
2267970	12/2009	EP	N/A
2854349	12/2014	EP	N/A
3772208	12/2023	EP	N/A
10-2011-0045459	12/2010	KR	N/A
2006/122024	12/2005	WO	N/A
2022/121461	12/2021	WO	N/A
2022/133026	12/2021	WO	N/A
2023/240360	12/2022	WO	N/A
2025/059187	12/2024	WO	N/A

OTHER PUBLICATIONS

“About Azure Key Vault certificates”, Retrieved from <https://learn.microsoft.com/en-us/azure/key-vault/certificates/about-certificates>, Feb. 8, 2023, pp. 1-8. cited by applicant

“About the Expressway”, Aug. 17, 2022. pp. 1-12. cited by applicant

“Automated certificate management for TLS certificates”, Retrieved from <https://docs.servicenow.com/en-US/bundle/utah-it-operations-management/page/product/discovery/concept/automated-cert-requests.html>, Retrieved on May 4, 2023, pp. 1-4. cited by applicant

“AWS Certificate Manager FAQs”, Retrieved from <https://aws.amazon.com/certificate-manager/faqs/>, Retrieved on Mar. 24, 2023, pp. 1-17. cited by applicant

“Azure Instance Metadata Service”, Retrieved from <https://learn.microsoft.com/en-us/azure/virtual-machines/instance-metadata-service?tabs=windows>, Mar. 15, 2023, pp. 1-42. cited by applicant

“Cisco Expressway Certificate Creation and Use Deployment Guide”, Feb. 23, 2021, pp. 10. cited by applicant

“Deploying the CA bundle iApp”, Retrieved from <https://www.f5.com/pdf/deployment-guides/f5-ca-bundle-dg.pdf>, Dec. 14, 2017, pp. 1-9. cited by applicant

“DigiCert Public Key Infrastructure (PKI) Platform”, 2019, pp. 15. cited by applicant

“Get started with Key Vault certificates”, Retrieved from <https://learn.microsoft.com/en-us/azure/key-vault/certificates/certificate-scenarios>, Retrieved on Feb. 1, 2023, pp. 1-6. cited by applicant

“High Availability using Patching and Rolling AP Upgrade on Cisco Catalyst 9800 Wireless

Controllers”, Copyright 2020, pp. 1-41. cited by applicant

“Manage Certificate Revocation Lists (CRLs)”, Jul. 23, 2021, pp. 1-4. cited by applicant

“PKI secrets engine”, Retrieved from <https://developer.hashicorp.com/vault/docs/secrets/pki>, Retrieved on May 4, 2023, pp. 1-3. cited by applicant

“Planning a certificate revocation list (CRL)”, Retrieved from <https://docs.aws.amazon.com/privateca/latest/userguide/crl-planning.html>, Retrieved on Jul. 28, 2023, pp. 11. cited by applicant

“Release app updates with staged rollouts”, Retrieved from <https://support.google.com/googleplay/android-developer/answer/6346149?hl=en#zippy=%2C%20release-a-staged-rollout-to-specific-countries>, Retrieved on Apr. 27, 2023, pp. 1-2. cited by applicant

“Release Your App Update in a Staged Rollout”, Retrieved from <https://developer.amazon.com/docs/app-submission/release-updates-in-staged-rollouts.html>, Retrieved on Apr. 27, 2023, pp. 1-18. cited by applicant

“Rotate Security Certificates”, Retrieved from <https://www.cockroachlabs.com/docs/stable/rotate-certificates>, Retrieved on May 4, 2023, pp. 1-6. cited by applicant

“Rotating the Root CA and Leaf Certificates”, Retrieved from <https://docs.pivotal.io/ops-manager/2-4/security/pcf-infrastructure/rotate-cas-and-leaf-certs.html>, Nov. 5, 2020, pp. 1-9. cited by applicant

“Staged upgrade”, Retrieved from <https://www.ibm.com/docs/en/order-management-sw/9.4.0?topic=migrating-staged-upgrade>, Mar. 2, 2021, pp. 1-3. cited by applicant

“Troubleshoot SSL certificates”, Retrieved from <https://cloud.google.com/load-balancing/docs/ssl-certificates/troubleshooting>, Retrieved on Mar. 24, 2023, pp. 1-8. cited by applicant

“Tutorial: Configure certificate auto-rotation in Key Vault”, Retrieved from <https://learn.microsoft.com/en-us/azure/key-vault/certificates/tutorial-rotate-certificates>, Feb. 27, 2023, pp. 1-6. cited by applicant

“Updating the CA bundle”, Retrieved from https://docs.openshift.com/container-platform/4.9/security/certificates/updating-ca-bundle.html#ca-bundle-understanding_updating-ca-bundle, Retrieved on Mar. 24, 2023, pp. 1-2. cited by applicant

“Updating your private CA”, Retrieved from <https://docs.aws.amazon.com/privateca/latest/userguide/PCAUpdateCA.html>, Retrieved on Mar. 24, 2023, pp. 1-4. cited by applicant

“Use self-managed SSL certificates”, Retrieved from <https://cloud.google.com/load-balancing/docs/ssl-certificates/self-managed-certs>, Aug. 15, 2023, pp. 13. cited by applicant

“VSphere Security”, vmware, Update 3, Mar. 21, 2023, pp. 1-426. cited by applicant

“Working with Hosts”, Retrieved from <https://docs.cloudstack.apache.org/projects/archived-cloudstack-administration/en/latest/hosts.html>, Retrieved on Mar. 24, 2023, pp. 1-7. cited by applicant

Atutxa et al., “Improving efficiency and security of IIoT communications using in-network validation of server certificate”, Computers in Industry, vol. 144, Jan. 2023, 103802, pp. 30. cited by applicant

Bigelow S.J., “Rolling deployment”, Retrieved from <https://www.techtarget.com/searchitoperations/definition/rolling-deployment>, Jan. 2023, pp. 4. cited by applicant

Este-Gracias S., “Rotate your CA seamlessly using a Vault PKI”, Retrieved from <https://sestegra.medium.com/rotate-your-ca-seamlessly-using-a-vault-pki-9262228b4afb> Sep. 29, 2022, pp. 1-49. cited by applicant

Ghanmi et al., “A Secure Data Storage in Multi-cloud Architecture Using Blowfish Encryption Algorithm”, Advanced Information Networking and Applications, Mar. 2022, pp. 398-408. cited by applicant

Jamal F., “Zero Trust for SSH—Secure One-click Server Access for Software Engineering Teams”, Retrieved from <https://www.banyansecurity.io/blog/zero-trust-for-ssh/>, Oct. 28, 2020, pp. 1-7. cited by applicant

Manjusha R. et al., “Secure Authentication and Access System for Cloud Computing Auditing Services Using Associated Digital Certificate”, Indian Journal of Science and Technology, vol. 8 (S7), Apr. 2015, pp. 220-227. cited by applicant

Nexthop Team, “Updated: Creating a Certificate Revocation List Distribution Point for Your Internal Certification Authority”, Retrieved from <https://techcommunity.microsoft.com/t5/skype-for-business-blog/updated-creating-a-certificate-revocation-list-distribution/ba-p/620691>, Dec. 17, 2012, pp. 10. cited by applicant

Rowley J., “Google's Moving Forward Together Proposals for Root CA Policy: Rotating ICAS More Frequently”, Retrieved from <https://www.digicert.com/blog/googles-moving-forward-together-proposals-for-root-ca-policy>, Mar. 22, 2023, pp. 1-11. cited by applicant

Subhayu, “Different Phases of a Certificate Lifecycle Management Process for a secure WPA2-Enterprise network”, Certificate Lifecycle Management Oct. 6, 2022, pp. 16. cited by applicant

Ylonen et al., “Security of Automated Access Management Using Secure Shell (SSH)”, NISTIR 7966 (Draft), Aug. 2014, pp. 43. cited by applicant

“What is Certificate Lifecycle Management”, Retrieved from <https://www.encryptionconsulting.com/different-phases-of-a-certificate-lifecycle-management-process/>, Aug. 1, 2024, pp. 1-12. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to digital certificates used by network entities to authenticate other network entities. In particular, the present disclosure relates to distributing certificate bundles, that include certificate authority (CA) certificates, to network entities associated with a computer network.

BACKGROUND

(2) A computer network, such as a virtual cloud network, includes network entities that communicate with one another. Communications between network entities may be performed in accordance with a security protocol whereby network entities authenticate one another by presenting a digital certificate. When a network entity presents a valid digital certificate to another network entity, the other network entity can trust that it is communicating with the network entity, as opposed to some unknown entity, based on a trust relationship with the CA that issued the digital certificate.

(3) A digital certificate may be issued to a network entity by a CA. The digital certificate may include a digital signature generated using a private key of the CA that issued the digital certificate. The digital signature can be validated using CA certificate of the CA that includes a public key corresponding to the private key. Various CA certificates that may be utilized to validate digital certificates issued to various network entities may be included in a certificate bundle. The certificate bundle may be distributed to network entities throughout a computer network, such as a virtual cloud network. For example, a certificate bundle may be installed when initializing the computer network and/or when initializing various network entities on the computer network.

Additionally, a certificate bundle may be updated or replaced from time-to-time, for example, to make new CA certificates available to the network entities and/or to remove old CA certificates. (4) The content of this background section should not be construed as prior art merely by virtue of its presence in this section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings. It should be noted that references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and they mean at least one. In the drawings:

- (2) FIGS. 1-4 are block diagrams illustrating patterns for implementing a cloud infrastructure as a service system in accordance with one or more embodiments;
- (3) FIG. 5 is a hardware system in accordance with one or more embodiments;
- (4) FIG. 6A illustrates features of an example virtual cloud network in accordance with one or more embodiments;
- (5) FIGS. 6B and 6C respectively illustrate an example certificate bundle distribution schedule in accordance with one or more embodiments;
- (6) FIG. 6D illustrates further features of an example virtual cloud network in accordance with one or more embodiments;
- (7) FIG. 7 is a flowchart that illustrates an example certificate bundle distribution process for distributing a set of CA certificates to network entities associated with a virtual cloud network in accordance with one or more embodiments; and
- (8) FIGS. 8A and 8B are flowcharts that illustrates example operations pertaining to distributing certificate bundles to network entities in accordance with one or more embodiments.

DETAILED DESCRIPTION

(9) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding. One or more embodiments may be practiced without these specific details. Features described in one embodiment may be combined with features described in a different embodiment. In some examples, well-known structures and devices are described with reference to a block diagram form in order to avoid unnecessarily obscuring the present invention. Asymmetric 1. GENERAL OVERVIEW 2. CLOUD COMPUTING TECHNOLOGY 3. COMPUTER SYSTEM 4. ARCHITECTURAL OVERVIEW 5. EXAMPLE CERTIFICATE BUNDLE DISTRIBUTION PROCESS 6. EXAMPLE CERTIFICATE BUNDLE DISTRIBUTION OPERATIONS 7. AUTHENTICATING NETWORK ENTITIES 8. MISCELLANEOUS; EXTENSIONS

1. General Overview

(10) One or more embodiments associated network entities of a computer network, such as a virtual cloud network, with particular certificate distribution groups based on a set of one or more partitioning criteria that includes a fault domain of the particular network entity. Additionally, the set of partitioning criteria may include a resource group of the particular network entity. A certificate bundle that includes a set of one or more CA certificates may be distributed to the certificate distribution groups in a series of release phases that respectively correspond to a particular certificate distribution group. By distributing a certificate bundle sequentially to a series of certificate distribution groups, the possibility of an unsuccessful distribution may be confined to a subset of network entities corresponding to the particular certificate distribution group. In one example, the subset of network entities in a particular certificate distribution group may be confined to a particular fault domain, and in the event of an unsuccessful distribution, other fault

domains may be unaffected. Additionally, or in the alternative, the subset of network entities may be confined to a particular resource group, and in the event of an unsuccessful distribution, other resource groups may be unaffected. Further, a successful distribution may be confirmed with respect to a particular certificate distribution group prior to proceeding with distribution of the certificate bundle to a next certificate distribution group in the sequence.

(11) One or more embodiments described in this Specification and/or recited in the claims may not be included in this General Overview section.

2. Cloud Computing Technology

(12) Infrastructure as a Service (IaaS) is an application of cloud computing technology. IaaS can be configured to provide virtualized computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (example services include billing software, monitoring software, logging software, load balancing software, clustering software, etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance.

(13) In some instances, IaaS customers may access resources and services through a wide area network (WAN), such as the Internet, and can use the cloud provider's services to install the remaining elements of an application stack. For example, the user can log in to the IaaS platform to create virtual machines (VMs), install operating systems (OSs) on each VM, deploy middleware such as databases, create storage buckets for workloads and backups, and even install enterprise software into that VM. Customers can then use the provider's services to perform various functions, including balancing network traffic, troubleshooting application issues, monitoring performance, managing disaster recovery, etc.

(14) In some cases, a cloud computing model will involve the participation of a cloud provider. The cloud provider may, but need not be, a third-party service that specializes in providing (e.g., offering, renting, selling) IaaS. An entity may also opt to deploy a private cloud, becoming its own provider of infrastructure services.

(15) In some examples, IaaS deployment is the process of implementing a new application, or a new version of an application, onto a prepared application server or other similar device. IaaS deployment may also include the process of preparing the server (e.g., installing libraries, daemons, etc.). The deployment process is often managed by the cloud provider, below the hypervisor layer (e.g., the servers, storage, network hardware, and virtualization). Thus, the customer may be responsible for handling (OS), middleware, and/or application deployment (e.g., on self-service virtual machines (e.g., that can be spun up on demand) or the like).

(16) In some examples, IaaS provisioning may refer to acquiring computers or virtual hosts for use, and even installing needed libraries or services on them. In most cases, deployment does not include provisioning, and the provisioning may need to be performed first.

(17) In some cases, there are challenges for IaaS provisioning. There is an initial challenge of provisioning the initial set of infrastructure. There is an additional challenge of evolving the existing infrastructure (e.g., adding new services, changing services, removing services, etc.) after the initial provisioning is completed. In some cases, these challenges may be addressed by enabling the configuration of the infrastructure to be defined declaratively. In other words, the infrastructure (e.g., what components are needed and how they interact) can be defined by one or more configuration files. Thus, the overall topology of the infrastructure (e.g., what resources depend on which, and how they each work together) can be described declaratively. In some instances, once the topology is defined, a workflow can be generated that creates and/or manages the different components described in the configuration files.

(18) In some examples, an infrastructure may have many interconnected elements. For example,

there may be one or more virtual private clouds (VPCs) (e.g., a potentially on-demand pool of configurable and/or shared computing resources), also known as a core network. In some examples, there may also be one or more inbound/outbound traffic group rules provisioned to define how the inbound and/or outbound traffic of the network will be set up and one or more virtual machines (VMs). Other infrastructure elements may also be provisioned, such as a load balancer, a database, or the like. As more and more infrastructure elements are desired and/or added, the infrastructure may incrementally evolve.

(19) In some instances, continuous deployment techniques may be employed to enable deployment of infrastructure code across various virtual computing environments. Additionally, the described techniques can enable infrastructure management within these environments. In some examples, service teams can write code that is desired to be deployed to one or more, but often many, different production environments (e.g., across various different geographic locations, sometimes spanning the entire world). In some embodiments, infrastructure and resources may be provisioned (manually, and/or using a provisioning tool) prior to deployment of code to be executed on the infrastructure. However, in some examples, the infrastructure on which the code will be deployed must first be set up. In some instances, the provisioning can be done manually, a provisioning tool may be utilized to provision the resources, and/or deployment tools may be utilized to deploy the code once the infrastructure is provisioned.

(20) FIG. 1 is a block diagram illustrating an example pattern of an IaaS architecture **100**, according to at least one embodiment. Service operators **102** can be communicatively coupled to a secure host tenancy **104** that can include a virtual cloud network (VCN) **106** and a secure host subnet **108**. In some examples, the service operators **102** may be using one or more client computing devices, which may be portable handheld devices (e.g., an iPhone®, cellular telephone, an iPad®, computing tablet, a personal digital assistant (PDA)) or wearable devices (e.g., a Google Glass® head mounted display), running software such as Microsoft Windows Mobile®, and/or a variety of mobile operating systems such as iOS, Windows Phone, Android, BlackBerry 8, Palm OS, and the like, and being Internet, e-mail, short message service (SMS), Blackberry®, or other communication protocol enabled. Alternatively, the client computing devices can be general purpose personal computers including, by way of example, personal computers and/or laptop computers running various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems. The client computing devices can be workstation computers running any of a variety of commercially-available UNIX® or UNIX-like operating systems, including without limitation the variety of GNU/Linux operating systems, such as for example, Google Chrome OS. Alternatively, or in addition, client computing devices may be any other electronic device, such as a thin-client computer, an Internet-enabled gaming system (e.g., a Microsoft Xbox gaming console with or without a Kinect® gesture input device), and/or a personal messaging device, capable of communicating over a network that can access the VCN **106** and/or the Internet.

(21) The VCN **106** can include a local peering gateway (LPG) **110** that can be communicatively coupled to a secure shell (SSH) VCN **112** via an LPG **110** contained in the SSH VCN **112**. The SSH VCN **112** can include an SSH subnet **114**, and the SSH VCN **112** can be communicatively coupled to a control plane VCN **116** via the LPG **110** contained in the control plane VCN **116**. Also, the SSH VCN **112** can be communicatively coupled to a data plane VCN **118** via an LPG **110**. The control plane VCN **116** and the data plane VCN **118** can be contained in a service tenancy **119** that can be owned and/or operated by the IaaS provider.

(22) The control plane VCN **116** can include a control plane demilitarized zone (DMZ) tier **120** that acts as a perimeter network (e.g., portions of a corporate network between the corporate intranet and external networks). The DMZ-based servers may have restricted responsibilities and help keep breaches contained. Additionally, the DMZ tier **120** can include one or more load balancer (LB) subnet(s) **122**, a control plane app tier **124** that can include app subnet(s) **126**, a control plane data tier **128** that can include database (DB) subnet(s) **130** (e.g., frontend DB subnet(s) and/or backend

DB subnet(s)). The LB subnet(s) **122** contained in the control plane DMZ tier **120** can be communicatively coupled to the app subnet(s) **126** contained in the control plane app tier **124** and an Internet gateway **134** that can be contained in the control plane VCN **116**, and the app subnet(s) **126** can be communicatively coupled to the DB subnet(s) **130** contained in the control plane data tier **128** and a service gateway **136** and a network address translation (NAT) gateway **138**. The control plane VCN **116** can include the service gateway **136** and the NAT gateway **138**.

(23) The control plane VCN **116** can include a data plane mirror app tier **140** that can include app subnet(s) **126**. The app subnet(s) **126** contained in the data plane mirror app tier **140** can include a virtual network interface controller (VNIC) **142** that can execute a compute instance **144**. The compute instance **144** can communicatively couple the app subnet(s) **126** of the data plane mirror app tier **140** to app subnet(s) **126** that can be contained in a data plane app tier **146**.

(24) The data plane VCN **118** can include the data plane app tier **146**, a data plane DMZ tier **148**, and a data plane data tier **150**. The data plane DMZ tier **148** can include LB subnet(s) **122** that can be communicatively coupled to the app subnet(s) **126** of the data plane app tier **146** and the Internet gateway **134** of the data plane VCN **118**. The app subnet(s) **126** can be communicatively coupled to the service gateway **136** of the data plane VCN **118** and the NAT gateway **138** of the data plane VCN **118**. The data plane data tier **150** can also include the DB subnet(s) **130** that can be communicatively coupled to the app subnet(s) **126** of the data plane app tier **146**.

(25) The Internet gateway **134** of the control plane VCN **116** and of the data plane VCN **118** can be communicatively coupled to a metadata management service **152** that can be communicatively coupled to public Internet **154**. Public Internet **154** can be communicatively coupled to the NAT gateway **138** of the control plane VCN **116** and of the data plane VCN **118**. The service gateway **136** of the control plane VCN **116** and of the data plane VCN **118** can be communicatively couple to cloud services **156**.

(26) In some examples, the service gateway **136** of the control plane VCN **116** or of the data plane VCN **118** can make application programming interface (API) calls to cloud services **156** without going through public Internet **154**. The API calls to cloud services **156** from the service gateway **136** can be one-way: the service gateway **136** can make API calls to cloud services **156**, and cloud services **156** can send requested data to the service gateway **136**. But, cloud services **156** may not initiate API calls to the service gateway **136**.

(27) In some examples, the secure host tenancy **104** can be directly connected to the service tenancy **119**, which may be otherwise isolated. The secure host subnet **108** can communicate with the SSH subnet **114** through an LPG **110** that may enable two-way communication over an otherwise isolated system. Connecting the secure host subnet **108** to the SSH subnet **114** may give the secure host subnet **108** access to other entities within the service tenancy **119**.

(28) The control plane VCN **116** may allow users of the service tenancy **119** to set up or otherwise provision desired resources. Desired resources provisioned in the control plane VCN **116** may be deployed or otherwise used in the data plane VCN **118**. In some examples, the control plane VCN **116** can be isolated from the data plane VCN **118**, and the data plane mirror app tier **140** of the control plane VCN **116** can communicate with the data plane app tier **146** of the data plane VCN **118** via VNICs **142** that can be contained in the data plane mirror app tier **140** and the data plane app tier **146**.

(29) In some examples, users of the system, or customers, can make requests, for example create, read, update, or delete (CRUD) operations, through public Internet **154** that can communicate the requests to the metadata management service **152**. The metadata management service **152** can communicate the request to the control plane VCN **116** through the Internet gateway **134**. The request can be received by the LB subnet(s) **122** contained in the control plane DMZ tier **120**. The LB subnet(s) **122** may determine that the request is valid, and in response to this determination, the LB subnet(s) **122** can transmit the request to app subnet(s) **126** contained in the control plane app tier **124**. If the request is validated and requires a call to public Internet **154**, the call to public

Internet **154** may be transmitted to the NAT gateway **138** that can make the call to public Internet **154**. Metadata that may be desired to be stored by the request can be stored in the DB subnet(s) **130**.

(30) In some examples, the data plane mirror app tier **140** can facilitate direct communication between the control plane VCN **116** and the data plane VCN **118**. For example, changes, updates, or other suitable modifications to configuration may be desired to be applied to the resources contained in the data plane VCN **118**. Via a VNIC **142**, the control plane VCN **116** can directly communicate with, and can thereby execute the changes, updates, or other suitable modifications to configuration to, resources contained in the data plane VCN **118**.

(31) In some embodiments, the control plane VCN **116** and the data plane VCN **118** can be contained in the service tenancy **119**. In this case, the user, or the customer, of the system may not own or operate either the control plane VCN **116** or the data plane VCN **118**. Instead, the IaaS provider may own or operate the control plane VCN **116** and the data plane VCN **118**, both of which may be contained in the service tenancy **119**. This embodiment can enable isolation of networks that may prevent users or customers from interacting with other users', or other customers', resources. Also, this embodiment may allow users or customers of the system to store databases privately without needing to rely on public Internet **154**, which may not have a desired level of threat prevention, for storage.

(32) In other embodiments, the LB subnet(s) **122** contained in the control plane VCN **116** can be configured to receive a signal from the service gateway **136**. In this embodiment, the control plane VCN **116** and the data plane VCN **118** may be configured to be called by a customer of the IaaS provider without calling public Internet **154**. Customers of the IaaS provider may desire this embodiment since database(s) that the customers use may be controlled by the IaaS provider and may be stored on the service tenancy **119**, which may be isolated from public Internet **154**.

(33) FIG. 2 is a block diagram illustrating another example pattern of an IaaS architecture **200**, according to at least one embodiment. Service operators **202** (e.g., service operators **102** of FIG. 1) can be communicatively coupled to a secure host tenancy **204** (e.g., the secure host tenancy **104** of FIG. 1) that can include a virtual cloud network (VCN) **206** (e.g., the VCN **106** of FIG. 1) and a secure host subnet **208** (e.g., the secure host subnet **108** of FIG. 1). The VCN **206** can include a local peering gateway (LPG) **210** (e.g., the LPG **110** of FIG. 1) that can be communicatively coupled to a secure shell (SSH) VCN **212** (e.g., the SSH VCN **112** of FIG. 1) via an LPG **110** contained in the SSH VCN **212**. The SSH VCN **212** can include an SSH subnet **214** (e.g., the SSH subnet **114** of FIG. 1), and the SSH VCN **212** can be communicatively coupled to a control plane VCN **216** (e.g., the control plane VCN **116** of FIG. 1) via an LPG **210** contained in the control plane VCN **216**. The control plane VCN **216** can be contained in a service tenancy **219** (e.g., the service tenancy **119** of FIG. 1), and the data plane VCN **218** (e.g., the data plane VCN **118** of FIG. 1) can be contained in a customer tenancy **221** that may be owned or operated by users, or customers, of the system.

(34) The control plane VCN **216** can include a control plane DMZ tier **220** (e.g., the control plane DMZ tier **120** of FIG. 1) that can include LB subnet(s) **222** (e.g., LB subnet(s) **122** of FIG. 1), a control plane app tier **224** (e.g., the control plane app tier **124** of FIG. 1) that can include app subnet(s) **226** (e.g., app subnet(s) **126** of FIG. 1), a control plane data tier **228** (e.g., the control plane data tier **128** of FIG. 1) that can include database (DB) subnet(s) **230** (e.g., similar to DB subnet(s) **130** of FIG. 1). The LB subnet(s) **222** contained in the control plane DMZ tier **220** can be communicatively coupled to the app subnet(s) **226** contained in the control plane app tier **224** and an Internet gateway **234** (e.g., the Internet gateway **134** of FIG. 1) that can be contained in the control plane VCN **216**, and the app subnet(s) **226** can be communicatively coupled to the DB subnet(s) **230** contained in the control plane data tier **228** and a service gateway **236** (e.g., the service gateway **136** of FIG. 1) and a network address translation (NAT) gateway **238** (e.g., the NAT gateway **138** of FIG. 1). The control plane VCN **216** can include the service gateway **236** and

the NAT gateway **238**.

(35) The control plane VCN **216** can include a data plane mirror app tier **240** (e.g., the data plane mirror app tier **140** of FIG. 1) that can include app subnet(s) **226**. The app subnet(s) **226** contained in the data plane mirror app tier **240** can include a virtual network interface controller (VNIC) **242** (e.g., the VNIC of **142**) that can execute a compute instance **244** (e.g., similar to the compute instance **144** of FIG. 1). The compute instance **244** can facilitate communication between the app subnet(s) **226** of the data plane mirror app tier **240** and the app subnet(s) **226** that can be contained in a data plane app tier **246** (e.g., the data plane app tier **146** of FIG. 1) via the VNIC **242** contained in the data plane mirror app tier **240** and the VNIC **242** contained in the data plane app tier **246**.

(36) The Internet gateway **234** contained in the control plane VCN **216** can be communicatively coupled to a metadata management service **252** (e.g., the metadata management service **152** of FIG. 1) that can be communicatively coupled to public Internet **254** (e.g., public Internet **154** of FIG. 1). Public Internet **254** can be communicatively coupled to the NAT gateway **238** contained in the control plane VCN **216**. The service gateway **236** contained in the control plane VCN **216** can be communicatively couple to cloud services **256** (e.g., cloud services **156** of FIG. 1).

(37) In some examples, the data plane VCN **218** can be contained in the customer tenancy **221**. In this case, the IaaS provider may provide the control plane VCN **216** for each customer, and the IaaS provider may, for each customer, set up a unique, compute instance **244** that is contained in the service tenancy **219**. Each compute instance **244** may allow communication between the control plane VCN **216**, contained in the service tenancy **219**, and the data plane VCN **218** that is contained in the customer tenancy **221**. The compute instance **244** may allow resources, that are provisioned in the control plane VCN **216** that is contained in the service tenancy **219**, to be deployed or otherwise used in the data plane VCN **218** that is contained in the customer tenancy **221**.

(38) In other examples, the customer of the IaaS provider may have databases that live in the customer tenancy **221**. In this example, the control plane VCN **216** can include the data plane mirror app tier **240** that can include app subnet(s) **226**. The data plane mirror app tier **240** can reside in the data plane VCN **218**, but the data plane mirror app tier **240** may not live in the data plane VCN **218**. That is, the data plane mirror app tier **240** may have access to the customer tenancy **221**, but the data plane mirror app tier **240** may not exist in the data plane VCN **218** or be owned or operated by the customer of the IaaS provider. The data plane mirror app tier **240** may be configured to make calls to the data plane VCN **218** but may not be configured to make calls to any entity contained in the control plane VCN **216**. The customer may desire to deploy or otherwise use resources in the data plane VCN **218** that are provisioned in the control plane VCN **216**, and the data plane mirror app tier **240** can facilitate the desired deployment, or other usage of resources, of the customer.

(39) In some embodiments, the customer of the IaaS provider can apply filters to the data plane VCN **218**. In this embodiment, the customer can determine what the data plane VCN **218** can access, and the customer may restrict access to public Internet **254** from the data plane VCN **218**. The IaaS provider may not be able to apply filters or otherwise control access of the data plane VCN **218** to any outside networks or databases. Applying filters and controls by the customer onto the data plane VCN **218**, contained in the customer tenancy **221**, can help isolate the data plane VCN **218** from other customers and from public Internet **254**.

(40) In some embodiments, cloud services **256** can be called by the service gateway **236** to access services that may not exist on public Internet **254**, on the control plane VCN **216**, or on the data plane VCN **218**. The connection between cloud services **256** and the control plane VCN **216** or the data plane VCN **218** may not be live or continuous. Cloud services **256** may exist on a different network owned or operated by the IaaS provider. Cloud services **256** may be configured to receive calls from the service gateway **236** and may be configured to not receive calls from public Internet **254**. Some cloud services **256** may be isolated from other cloud services **256**, and the control plane

VCN **216** may be isolated from cloud services **256** that may not be in the same region as the control plane VCN **216**. For example, the control plane VCN **216** may be located in “Region 1,” and cloud service “Deployment 1,” may be located in Region 1 and in “Region 2.” If a call to Deployment 1 is made by the service gateway **236** contained in the control plane VCN **216** located in Region 1, the call may be transmitted to Deployment 1 in Region 1. In this example, the control plane VCN **216**, or Deployment 1 in Region 1, may not be communicatively coupled to, or otherwise in communication with, Deployment 1 in Region 2.

(41) FIG. **3** is a block diagram illustrating another example pattern of an IaaS architecture **300**, according to at least one embodiment. Service operators **302** (e.g., service operators **102** of FIG. **1**) can be communicatively coupled to a secure host tenancy **304** (e.g., the secure host tenancy **104** of FIG. **1**) that can include a virtual cloud network (VCN) **306** (e.g., the VCN **106** of FIG. **1**) and a secure host subnet **308** (e.g., the secure host subnet **108** of FIG. **1**). The VCN **306** can include an LPG **310** (e.g., the LPG **110** of FIG. **1**) that can be communicatively coupled to an SSH VCN **312** (e.g., the SSH VCN **112** of FIG. **1**) via an LPG **310** contained in the SSH VCN **312**. The SSH VCN **312** can include an SSH subnet **314** (e.g., the SSH subnet **114** of FIG. **1**), and the SSH VCN **312** can be communicatively coupled to a control plane VCN **316** (e.g., the control plane VCN **116** of FIG. **1**) via an LPG **310** contained in the control plane VCN **316** and to a data plane VCN **318** (e.g., the data plane VCN **118** of FIG. **1**) via an LPG **310** contained in the data plane VCN **318**. The control plane VCN **316** and the data plane VCN **318** can be contained in a service tenancy **319** (e.g., the service tenancy **119** of FIG. **1**).

(42) The control plane VCN **316** can include a control plane DMZ tier **320** (e.g., the control plane DMZ tier **120** of FIG. **1**) that can include load balancer (LB) subnet(s) **322** (e.g., LB subnet(s) **122** of FIG. **1**), a control plane app tier **324** (e.g., the control plane app tier **124** of FIG. **1**) that can include app subnet(s) **326** (e.g., similar to app subnet(s) **126** of FIG. **1**), a control plane data tier **328** (e.g., the control plane data tier **128** of FIG. **1**) that can include DB subnet(s) **330**. The LB subnet(s) **322** contained in the control plane DMZ tier **320** can be communicatively coupled to the app subnet(s) **326** contained in the control plane app tier **324** and to an Internet gateway **334** (e.g., the Internet gateway **134** of FIG. **1**) that can be contained in the control plane VCN **316**, and the app subnet(s) **326** can be communicatively coupled to the DB subnet(s) **330** contained in the control plane data tier **328** and to a service gateway **336** (e.g., the service gateway of FIG. **1**) and a network address translation (NAT) gateway **338** (e.g., the NAT gateway **138** of FIG. **1**). The control plane VCN **316** can include the service gateway **336** and the NAT gateway **338**.

(43) The data plane VCN **318** can include a data plane app tier **346** (e.g., the data plane app tier **146** of FIG. **1**), a data plane DMZ tier **348** (e.g., the data plane DMZ tier **148** of FIG. **1**), and a data plane data tier **350** (e.g., the data plane data tier **150** of FIG. **1**). The data plane DMZ tier **348** can include LB subnet(s) **322** that can be communicatively coupled to trusted app subnet(s) **360** and untrusted app subnet(s) **362** of the data plane app tier **346** and the Internet gateway **334** contained in the data plane VCN **318**. The trusted app subnet(s) **360** can be communicatively coupled to the service gateway **336** contained in the data plane VCN **318**, the NAT gateway **338** contained in the data plane VCN **318**, and DB subnet(s) **330** contained in the data plane data tier **350**. The untrusted app subnet(s) **362** can be communicatively coupled to the service gateway **336** contained in the data plane VCN **318** and DB subnet(s) **330** contained in the data plane data tier **350**. The data plane data tier **350** can include DB subnet(s) **330** that can be communicatively coupled to the service gateway **336** contained in the data plane VCN **318**.

(44) The untrusted app subnet(s) **362** can include one or more primary VNICS **364(1)-(N)** that can be communicatively coupled to tenant virtual machines (VMs) **366(1)-(N)**. Each tenant VM **366(1)-(N)** can be communicatively coupled to a respective app subnet **367(1)-(N)** that can be contained in respective container egress VCNs **368(1)-(N)** that can be contained in respective customer tenancies **380(1)-(N)**. Respective secondary VNICS **372(1)-(N)** can facilitate communication between the untrusted app subnet(s) **362** contained in the data plane VCN **318** and

the app subnet contained in the container egress VCNs **368(1)-(N)**. Each container egress VCNs **368(1)-(N)** can include a NAT gateway **338** that can be communicatively coupled to public Internet **354** (e.g., public Internet **154** of FIG. 1).

(45) The Internet gateway **334** contained in the control plane VCN **316** and contained in the data plane VCN **318** can be communicatively coupled to a metadata management service **352** (e.g., the metadata management service **152** of FIG. 1) that can be communicatively coupled to public Internet **354**. Public Internet **354** can be communicatively coupled to the NAT gateway **338** contained in the control plane VCN **316** and contained in the data plane VCN **318**. The service gateway **336** contained in the control plane VCN **316** and contained in the data plane VCN **318** can be communicatively couple to cloud services **356**.

(46) In some embodiments, the data plane VCN **318** can be integrated with customer tenancies **380**. This integration can be useful or desirable for customers of the IaaS provider in some cases such as a case that may desire support when executing code. The customer may provide code to run that may be destructive, may communicate with other customer resources, or may otherwise cause undesirable effects. In response to this, the IaaS provider may determine whether to run code given to the IaaS provider by the customer.

(47) In some examples, the customer of the IaaS provider may grant temporary network access to the IaaS provider and request a function to be attached to the data plane app tier **346**. Code to run the function may be executed in the VMs **366(1)-(N)**, and the code may not be configured to run anywhere else on the data plane VCN **318**. Each VM **366(1)-(N)** may be connected to one customer tenancy **380**. Respective containers **381(1)-(N)** contained in the VMs **366(1)-(N)** may be configured to run the code. In this case, there can be a dual isolation (e.g., the containers **381(1)-(N)** running code, where the containers **381(1)-(N)** may be contained in at least the VM **366(1)-(N)** that are contained in the untrusted app subnet(s) **362**), which may help prevent incorrect or otherwise undesirable code from damaging the network of the IaaS provider or from damaging a network of a different customer. The containers **381(1)-(N)** may be communicatively coupled to the customer tenancy **380** and may be configured to transmit or receive data from the customer tenancy **380**. The containers **381(1)-(N)** may not be configured to transmit or receive data from any other entity in the data plane VCN **318**. Upon completion of running the code, the IaaS provider may kill or otherwise dispose of the containers **381(1)-(N)**.

(48) In some embodiments, the trusted app subnet(s) **360** may run code that may be owned or operated by the IaaS provider. In this embodiment, the trusted app subnet(s) **360** may be communicatively coupled to the DB subnet(s) **330** and be configured to execute CRUD operations in the DB subnet(s) **330**. The untrusted app subnet(s) **362** may be communicatively coupled to the DB subnet(s) **330**, but in this embodiment, the untrusted app subnet(s) may be configured to execute read operations in the DB subnet(s) **330**. The containers **381(1)-(N)** that can be contained in the VM **366(1)-(N)** of each customer and that may run code from the customer may not be communicatively coupled with the DB subnet(s) **330**.

(49) In other embodiments, the control plane VCN **316** and the data plane VCN **318** may not be directly communicatively coupled. In this embodiment, there may be no direct communication between the control plane VCN **316** and the data plane VCN **318**. However, communication can occur indirectly through at least one method. An LPG **310** may be established by the IaaS provider that can facilitate communication between the control plane VCN **316** and the data plane VCN **318**. In another example, the control plane VCN **316** or the data plane VCN **318** can make a call to cloud services **356** via the service gateway **336**. For example, a call to cloud services **356** from the control plane VCN **316** can include a request for a service that can communicate with the data plane VCN **318**.

(50) FIG. 4 is a block diagram illustrating another example pattern of an IaaS architecture **400**, according to at least one embodiment. Service operators **402** (e.g., service operators **102** of FIG. 1) can be communicatively coupled to a secure host tenancy **404** (e.g., the secure host tenancy **104** of

FIG. 1) that can include a virtual cloud network (VCN) **406** (e.g., the VCN **106** of FIG. 1) and a secure host subnet **408** (e.g., the secure host subnet **108** of FIG. 1). The VCN **406** can include an LPG **410** (e.g., the LPG **110** of FIG. 1) that can be communicatively coupled to an SSH VCN **412** (e.g., the SSH VCN **112** of FIG. 1) via an LPG **410** contained in the SSH VCN **412**. The SSH VCN **412** can include an SSH subnet **414** (e.g., the SSH subnet **114** of FIG. 1), and the SSH VCN **412** can be communicatively coupled to a control plane VCN **416** (e.g., the control plane VCN **116** of FIG. 1) via an LPG **410** contained in the control plane VCN **416** and to a data plane VCN **418** (e.g., the data plane VCN **118** of FIG. 1) via an LPG **410** contained in the data plane VCN **418**. The control plane VCN **416** and the data plane VCN **418** can be contained in a service tenancy **419** (e.g., the service tenancy **119** of FIG. 1).

(51) The control plane VCN **416** can include a control plane DMZ tier **420** (e.g., the control plane DMZ tier **120** of FIG. 1) that can include LB subnet(s) **422** (e.g., LB subnet(s) **122** of FIG. 1), a control plane app tier **424** (e.g., the control plane app tier **124** of FIG. 1) that can include app subnet(s) **426** (e.g., app subnet(s) **126** of FIG. 1), a control plane data tier **428** (e.g., the control plane data tier **128** of FIG. 1) that can include DB subnet(s) **430** (e.g., DB subnet(s) **330** of FIG. 3). The LB subnet(s) **422** contained in the control plane DMZ tier **420** can be communicatively coupled to the app subnet(s) **426** contained in the control plane app tier **424** and to an Internet gateway **434** (e.g., the Internet gateway **134** of FIG. 1) that can be contained in the control plane VCN **416**, and the app subnet(s) **426** can be communicatively coupled to the DB subnet(s) **430** contained in the control plane data tier **428** and to a service gateway **436** (e.g., the service gateway of FIG. 1) and a network address translation (NAT) gateway **438** (e.g., the NAT gateway **138** of FIG. 1). The control plane VCN **416** can include the service gateway **436** and the NAT gateway **438**.

(52) The data plane VCN **418** can include a data plane app tier **446** (e.g., the data plane app tier **146** of FIG. 1), a data plane DMZ tier **448** (e.g., the data plane DMZ tier **148** of FIG. 1), and a data plane data tier **450** (e.g., the data plane data tier **150** of FIG. 1). The data plane DMZ tier **448** can include LB subnet(s) **422** that can be communicatively coupled to trusted app subnet(s) **460** (e.g., trusted app subnet(s) **360** of FIG. 3) and untrusted app subnet(s) **462** (e.g., untrusted app subnet(s) **362** of FIG. 3) of the data plane app tier **446** and the Internet gateway **434** contained in the data plane VCN **418**. The trusted app subnet(s) **460** can be communicatively coupled to the service gateway **436** contained in the data plane VCN **418**, the NAT gateway **438** contained in the data plane VCN **418**, and DB subnet(s) **430** contained in the data plane data tier **450**. The untrusted app subnet(s) **462** can be communicatively coupled to the service gateway **436** contained in the data plane VCN **418** and DB subnet(s) **430** contained in the data plane data tier **450**. The data plane data tier **450** can include DB subnet(s) **430** that can be communicatively coupled to the service gateway **436** contained in the data plane VCN **418**.

(53) The untrusted app subnet(s) **462** can include primary VNICs **464(1)-(N)** that can be communicatively coupled to tenant virtual machines (VMs) **466(1)-(N)** residing within the untrusted app subnet(s) **462**. Each tenant VM **466(1)-(N)** can run code in a respective container **467(1)-(N)**, and be communicatively coupled to an app subnet **426** that can be contained in a data plane app tier **446** that can be contained in a container egress VCN **468**. Respective secondary VNICs **472(1)-(N)** can facilitate communication between the untrusted app subnet(s) **462** contained in the data plane VCN **418** and the app subnet contained in the container egress VCN **468**. The container egress VCN can include a NAT gateway **438** that can be communicatively coupled to public Internet **454** (e.g., public Internet **154** of FIG. 1).

(54) The Internet gateway **434** contained in the control plane VCN **416** and contained in the data plane VCN **418** can be communicatively coupled to a metadata management service **452** (e.g., the metadata management service **152** of FIG. 1) that can be communicatively coupled to public Internet **454**. Public Internet **454** can be communicatively coupled to the NAT gateway **438** contained in the control plane VCN **416** and contained in the data plane VCN **418**. The service

gateway **436** contained in the control plane VCN **416** and contained in the data plane VCN **418** can be communicatively couple to cloud services **456**.

(55) In some examples, the pattern illustrated by the architecture of block diagram **400** of FIG. **4** may be considered an exception to the pattern illustrated by the architecture of block diagram **300** of FIG. **3** and may be desirable for a customer of the IaaS provider if the IaaS provider cannot directly communicate with the customer (e.g., a disconnected region). The respective containers **467(1)-(N)** that are contained in the VMs **466(1)-(N)** for each customer can be accessed in real-time by the customer. The containers **467(1)-(N)** may be configured to make calls to respective secondary VNICs **472(1)-(N)** contained in app subnet(s) **426** of the data plane app tier **446** that can be contained in the container egress VCN **468**. The secondary VNICs **472(1)-(N)** can transmit the calls to the NAT gateway **438** that may transmit the calls to public Internet **454**. In this example, the containers **467(1)-(N)** that can be accessed in real-time by the customer can be isolated from the control plane VCN **416** and can be isolated from other entities contained in the data plane VCN **418**. The containers **467(1)-(N)** may also be isolated from resources from other customers.

(56) In other examples, the customer can use the containers **467(1)-(N)** to call cloud services **456**. In this example, the customer may run code in the containers **467(1)-(N)** that requests a service from cloud services **456**. The containers **467(1)-(N)** can transmit this request to the secondary VNICs **472(1)-(N)** that can transmit the request to the NAT gateway that can transmit the request to public Internet **454**. Public Internet **454** can transmit the request to LB subnet(s) **422** contained in the control plane VCN **416** via the Internet gateway **434**. In response to determining the request is valid, the LB subnet(s) can transmit the request to app subnet(s) **426** that can transmit the request to cloud services **456** via the service gateway **436**.

(57) It should be appreciated that IaaS architectures **100**, **200**, **300**, **400** depicted in the figures may have other components than those depicted. Further, the embodiments shown in the figures are only some examples of a cloud infrastructure system that may incorporate an embodiment of the disclosure. In some other embodiments, the IaaS systems may have more or fewer components than shown in the figures, may combine two or more components, or may have a different configuration or arrangement of components.

(58) In certain embodiments, the IaaS systems described herein may include a suite of applications, middleware, and database service offerings that are delivered to a customer in a self-service, subscription-based, elastically scalable, reliable, highly available, and secure manner. An example of such an IaaS system is the Oracle Cloud Infrastructure (OCI) provided by the present assignee.

(59) In one or more embodiments, a computer network provides connectivity among a set of nodes. The nodes may be local to and/or remote from each other. The nodes are connected by a set of links. Examples of links include a coaxial cable, an unshielded twisted cable, a copper cable, an optical fiber, and a virtual link.

(60) A subset of nodes implements the computer network. Examples of such nodes include a switch, a router, a firewall, and a network address translator (NAT). Another subset of nodes uses the computer network. Such nodes (also referred to as “hosts”) may execute a client process and/or a server process. A client process makes a request for a computing service (such as, execution of a particular application, and/or storage of a particular amount of data). A server process responds by executing the requested service and/or returning corresponding data.

(61) A computer network may be a physical network, including physical nodes connected by physical links. A physical node is any digital device. A physical node may be a function-specific hardware device, such as a hardware switch, a hardware router, a hardware firewall, and a hardware NAT. Additionally or alternatively, a physical node may be a generic machine that is configured to execute various virtual machines and/or applications performing respective functions. A physical link is a physical medium connecting two or more physical nodes. Examples of links include a coaxial cable, an unshielded twisted cable, a copper cable, and an optical fiber.

(62) A computer network may be an overlay network. An overlay network is a logical network

implemented on top of another network (such as, a physical network). Each node in an overlay network corresponds to a respective node in the underlying network. Hence, each node in an overlay network is associated with both an overlay address (to address to the overlay node) and an underlay address (to address the underlay node that implements the overlay node). An overlay node may be a digital device and/or a software process (such as, a virtual machine, an application instance, or a thread) A link that connects overlay nodes is implemented as a tunnel through the underlying network. The overlay nodes at either end of the tunnel treat the underlying multi-hop path between them as a single logical link. Tunneling is performed through encapsulation and decapsulation.

(63) In an embodiment, a client may be local to and/or remote from a computer network. The client may access the computer network over other computer networks, such as a private network or the Internet. The client may communicate requests to the computer network using a communications protocol, such as Hypertext Transfer Protocol (HTTP). The requests are communicated through an interface, such as a client interface (such as a web browser), a program interface, or an application programming interface (API).

(64) In an embodiment, a computer network provides connectivity between clients and network resources. Network resources include hardware and/or software configured to execute server processes. Examples of network resources include a processor, a data storage, a virtual machine, a container, and/or a software application. Network resources are shared amongst multiple clients. Clients request computing services from a computer network independently of each other. Network resources are dynamically assigned to the requests and/or clients on an on-demand basis. Network resources assigned to each request and/or client may be scaled up or down based on, for example, (a) the computing services requested by a particular client, (b) the aggregated computing services requested by a particular tenant, and/or (c) the aggregated computing services requested of the computer network. Such a computer network may be referred to as a “cloud network.”

(65) In an embodiment, a service provider provides a cloud network to one or more end users. Various service models may be implemented by the cloud network, including but not limited to Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), and Infrastructure-as-a-Service (IaaS). In SaaS, a service provider provides end users the capability to use the service provider's applications, which are executing on the network resources. In PaaS, the service provider provides end users the capability to deploy custom applications onto the network resources. The custom applications may be created using programming languages, libraries, services, and tools supported by the service provider. In IaaS, the service provider provides end users the capability to provision processing, storage, networks, and other fundamental computing resources provided by the network resources. Any arbitrary applications, including an operating system, may be deployed on the network resources.

(66) In an embodiment, various deployment models may be implemented by a computer network, including but not limited to a private cloud, a public cloud, and a hybrid cloud. In a private cloud, network resources are provisioned for exclusive use by a particular group of one or more entities (the term “entity” as used herein refers to a corporation, organization, person, or other entity). The network resources may be local to and/or remote from the premises of the particular group of entities. In a public cloud, cloud resources are provisioned for multiple entities that are independent from each other (also referred to as “tenants” or “customers”). The computer network and the network resources thereof are accessed by clients corresponding to different tenants. Such a computer network may be referred to as a “multi-tenant computer network.” Several tenants may use a same particular network resource at different times and/or at the same time. The network resources may be local to and/or remote from the premises of the tenants. In a hybrid cloud, a computer network comprises a private cloud and a public cloud. An interface between the private cloud and the public cloud allows for data and application portability. Data stored at the private cloud and data stored at the public cloud may be exchanged through the interface. Applications

implemented at the private cloud and applications implemented at the public cloud may have dependencies on each other. A call from an application at the private cloud to an application at the public cloud (and vice versa) may be executed through the interface.

(67) In an embodiment, tenants of a multi-tenant computer network are independent of each other. For example, a business or operation of one tenant may be separate from a business or operation of another tenant. Different tenants may demand different network requirements for the computer network. Examples of network requirements include processing speed, amount of data storage, security requirements, performance requirements, throughput requirements, latency requirements, resiliency requirements, Quality of Service (QoS) requirements, tenant isolation, and/or consistency. The same computer network may need to implement different network requirements demanded by different tenants.

(68) In one or more embodiments, in a multi-tenant computer network, tenant isolation is implemented to ensure that the applications and/or data of different tenants are not shared with each other. Various tenant isolation approaches may be used.

(69) In an embodiment, each tenant is associated with a tenant ID. Each network resource of the multi-tenant computer network is tagged with a tenant ID. A tenant is permitted access to a particular network resource only if the tenant and the particular network resources are associated with a same tenant ID.

(70) In an embodiment, each tenant is associated with a tenant ID. Each application, implemented by the computer network, is tagged with a tenant ID. Additionally or alternatively, each data structure and/or dataset, stored by the computer network, is tagged with a tenant ID. A tenant is permitted access to a particular application, data structure, and/or dataset only if the tenant and the particular application, data structure, and/or dataset are associated with a same tenant ID.

(71) As an example, each database implemented by a multi-tenant computer network may be tagged with a tenant ID. Only a tenant associated with the corresponding tenant ID may access data of a particular database. As another example, each entry in a database implemented by a multi-tenant computer network may be tagged with a tenant ID. Only a tenant associated with the corresponding tenant ID may access data of a particular entry. However, the database may be shared by multiple tenants.

(72) In an embodiment, a subscription list indicates which tenants have authorization to access which applications. For each application, a list of tenant IDs of tenants authorized to access the application is stored. A tenant is permitted access to a particular application only if the tenant ID of the tenant is included in the subscription list corresponding to the particular application.

(73) In an embodiment, network resources (such as digital devices, virtual machines, application instances, and threads) corresponding to different tenants are isolated to tenant-specific overlay networks maintained by the multi-tenant computer network. As an example, packets from any source device in a tenant overlay network may only be transmitted to other devices within the same tenant overlay network. Encapsulation tunnels are used to prohibit any transmissions from a source device on a tenant overlay network to devices in other tenant overlay networks. Specifically, the packets, received from the source device, are encapsulated within an outer packet. The outer packet is transmitted from a first encapsulation tunnel endpoint (in communication with the source device in the tenant overlay network) to a second encapsulation tunnel endpoint (in communication with the destination device in the tenant overlay network). The second encapsulation tunnel endpoint decapsulates the outer packet to obtain the original packet transmitted by the source device. The original packet is transmitted from the second encapsulation tunnel endpoint to the destination device in the same particular overlay network.

3. Computer System

(74) FIG. 5 illustrates an example computer system **500**, in which various embodiments may be implemented. The system **500** may be used to implement any of the computer systems described above. As shown in FIG. 5, computer system **500** includes a processing unit **504** that communicates

with a number of peripheral subsystems via a bus subsystem **502**. These peripheral subsystems may include a processing acceleration unit **506**, an I/O subsystem **508**, a storage subsystem **518** and a communications subsystem **524**. Storage subsystem **518** includes tangible computer-readable storage media **522** and a system memory **510**.

(75) Bus subsystem **502** provides a mechanism for letting the various components and subsystems of computer system **500** communicate with each other as intended. Although bus subsystem **502** is shown schematically as a single bus, alternative embodiments of the bus subsystem may utilize multiple buses. Bus subsystem **502** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which can be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard.

(76) Processing unit **504**, which can be implemented as one or more integrated circuits (e.g., a conventional microprocessor or microcontroller), controls the operation of computer system **500**. One or more processors may be included in processing unit **504**. These processors may include single core or multicore processors. In certain embodiments, processing unit **504** may be implemented as one or more independent processing units **532** and/or **534** with single or multicore processors included in each processing unit. In other embodiments, processing unit **504** may also be implemented as a quad-core processing unit formed by integrating two dual-core processors into a single chip.

(77) In various embodiments, processing unit **504** can execute a variety of programs in response to program code and can maintain multiple concurrently executing programs or processes. At any given time, some or all of the program code to be executed can be resident in processing unit **504** and/or in storage subsystem **518**. Through suitable programming, processing unit **504** can provide various functionalities described above. Computer system **500** may additionally include a processing acceleration unit **506**, which can include a digital signal processor (DSP), a special-purpose processor, and/or the like.

(78) I/O subsystem **508** may include user interface input devices and user interface output devices. User interface input devices may include a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may include, for example, motion sensing and/or gesture recognition devices such as the Microsoft Kinect® motion sensor that enables users to control and interact with an input device, such as the Microsoft Xbox® 360 game controller, through a natural user interface using gestures and spoken commands. User interface input devices may also include eye gesture recognition devices such as the Google Glass® blink detector that detects eye activity (e.g., ‘blinking’ while taking pictures and/or making a menu selection) from users and transforms the eye gestures as input into an input device (e.g., Google Glass®). Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems (e.g., Siri® navigator), through voice commands.

(79) User interface input devices may also include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments and

the like.

(80) User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. In general, use of the term “output device” is intended to include all possible types of devices and mechanisms for outputting information from computer system **500** to a user or other computer. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

(81) Computer system **500** may comprise a storage subsystem **518** that provides a tangible non-transitory computer-readable storage medium for storing software and data constructs that provide the functionality of the embodiments described in this disclosure. The software can include programs, code modules, instructions, scripts, etc., that when executed by one or more cores or processors of processing unit **504** provide the functionality described above. Storage subsystem **518** may also provide a repository for storing data used in accordance with the present disclosure.

(82) As depicted in the example in FIG. 5, storage subsystem **518** can include various components including a system memory **510**, computer-readable storage media **522**, and a computer readable storage media reader **520**. System memory **510** may store program instructions, such as application programs **512**, that are loadable and executable by processing unit **504**. System memory **510** may also store data, such as program data **514**, that is used during the execution of the instructions and/or data that is generated during the execution of the program instructions. Various different kinds of programs may be loaded into system memory **510** including but not limited to client applications, Web browsers, mid-tier applications, relational database management systems (RDBMS), virtual machines, containers, etc.

(83) System memory **510** may also store an operating system **516**. Examples of operating system **516** may include various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems, a variety of commercially-available UNIX® or UNIX-like operating systems (including without limitation the variety of GNU/Linux operating systems, the Google Chrome® OS, and the like) and/or mobile operating systems such as iOS, Windows® Phone, Android® OS, BlackBerry® OS, and Palm® OS operating systems. In certain implementations where computer system **500** executes one or more virtual machines, the virtual machines along with their guest operating systems (GOSs) may be loaded into system memory **510** and executed by one or more processors or cores of processing unit **504**.

(84) System memory **510** can come in different configurations depending upon the type of computer system **500**. For example, system memory **510** may be volatile memory (such as random access memory (RAM)) and/or non-volatile memory (such as read-only memory (ROM), flash memory, etc.) Different types of RAM configurations may be provided including a static random access memory (SRAM), a dynamic random access memory (DRAM), and others. In some implementations, system memory **510** may include a basic input/output system (BIOS) containing basic routines that help to transfer information between elements within computer system **500**, such as during start-up.

(85) Computer-readable storage media **522** may represent remote, local, fixed, and/or removable storage devices plus storage media for temporarily and/or more permanently containing, storing, computer-readable information for use by computer system **500** including instructions executable by processing unit **504** of computer system **500**.

(86) Computer-readable storage media **522** can include any appropriate media known or used in the art, including storage media and communication media, such as but not limited to, volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage and/or transmission of information. This can include tangible computer-readable storage

media such as RAM, ROM, electronically programmable ROM (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disk (DVD), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other tangible computer readable media.

(87) By way of example, computer-readable storage media **522** may include a hard disk drive that reads from or writes to non-removable, nonvolatile magnetic media, a magnetic disk drive that reads from or writes to a removable, nonvolatile magnetic disk, and an optical disk drive that reads from or writes to a removable, nonvolatile optical disk such as a CD ROM, DVD, and Blu-Ray® disk, or other optical media. Computer-readable storage media **522** may include, but is not limited to, Zip® drives, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **522** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magnetoresistive RAM (MRAM) SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs. The disk drives and their associated computer-readable media may provide non-volatile storage of computer-readable instructions, data structures, program modules, and other data for computer system **500**.

(88) Machine-readable instructions executable by one or more processors or cores of processing unit **504** may be stored on a non-transitory computer-readable storage medium. A non-transitory computer-readable storage medium can include physically tangible memory or storage devices that include volatile memory storage devices and/or non-volatile storage devices. Examples of non-transitory computer-readable storage medium include magnetic storage media (e.g., disk or tapes), optical storage media (e.g., DVDs, CDs), various types of RAM, ROM, or flash memory, hard drives, floppy drives, detachable memory drives (e.g., USB drives), or other type of storage device.

(89) Communications subsystem **524** provides an interface to other computer systems and networks. Communications subsystem **524** serves as an interface for receiving data from and transmitting data to other systems from computer system **500**. For example, communications subsystem **524** may enable computer system **500** to connect to one or more devices via the Internet. In some embodiments communications subsystem **524** can include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G or EDGE (enhanced data rates for global evolution), WiFi (IEEE 802.11 family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some embodiments communications subsystem **524** can provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

(90) In some embodiments, communications subsystem **524** may also receive input communication in the form of structured and/or unstructured data feeds **526**, event streams **528**, event updates **530**, and the like on behalf of one or more users who may use computer system **500**.

(91) By way of example, communications subsystem **524** may be configured to receive data feeds **526** in real-time from users of social networks and/or other communication services such as Twitter® feeds, Facebook® updates, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

(92) Additionally, communications subsystem **524** may also be configured to receive data in the form of continuous data streams, which may include event streams **528** of real-time events and/or event updates **530**, that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), clickstream analysis tools, automobile traffic monitoring, and the like.

(93) Communications subsystem **524** may also be configured to output the structured and/or unstructured data feeds **526**, event streams **528**, event updates **530**, and the like to one or more

databases that may be in communication with one or more streaming data source computers coupled to computer system **500**.

(94) Computer system **500** can be one of various types, including a handheld portable device (e.g., an iPhone® cellular phone, an iPad® computing tablet, a PDA), a wearable device (e.g., a Google Glass® head mounted display), a PC, a workstation, a mainframe, a kiosk, a server rack, or any other data processing system.

(95) Due to the ever-changing nature of computers and networks, the description of computer system **500** depicted in FIG. **5** is intended only as a specific example. Many other configurations having more or fewer components than the system depicted in FIG. **5** are possible. For example, customized hardware might also be used and/or particular elements might be implemented in hardware, firmware, software (including applets), or a combination. Further, connection to other computing devices, such as network input/output devices, may be employed. Based on the disclosure and teachings provided herein, a person of ordinary skill in the art will appreciate other ways and/or methods to implement the various embodiments.

4. Architectural Overview

(96) Referring now to FIGS. **6A-6D**, an example system **600** is described in accordance with one or more embodiments. The system **600** described with reference to FIGS. **6A-6D** may perform operations associated with distributing certificate bundles to network entities as described herein.

A. Example Virtual Cloud Network Architecture

(97) As shown in FIG. **6A**, the system **600** may include at least one virtual cloud network **601**. A virtual cloud network **601** may include a distribution service **602**, a certificate repository **604**, a distribution schedule repository **606**, and a plurality of network entities **608**. The distribution service **602** may distribute certificate bundles to the plurality of network entities **608**. The certificate bundles distributed to the network entities **608** may be stored in the certificate repository **604**. The distribution of the certificate bundles may be performed in accordance with a distribution schedule. The distribution schedule may be stored in the distribution schedule repository **606**.

(98) The distribution service **602** may include a distribution schedule generator **610** and a certificate transmitting engine **612**. The distribution schedule generator **610** may generate one or more distribution schedules for transmitting one or more certificate bundles to the plurality of network entities **608**. The certificate transmitting engine **612** may transmit certificate bundles to the plurality of network entities, for example, in accordance with a distribution schedule. The certificate transmitting engine **612** may transmit a certificate bundle to the plurality of network entities **608** in accordance with a push regime or a pull regime. In one example, for a push regime, the certificate transmitting engine **612** may transmit a certificate bundle to a plurality of network entities **608** responsive to determining a trigger condition. The trigger condition may include a new certificate bundle becoming available in the certificate repository **604**. Additionally, or in the alternative, the trigger condition may include a command in an executable file. In one example, for a pull regime, the certificate transmitting engine **612** may transmit a certificate bundle to one or more network entities **608** responsive to a request, such as from a particular network entity **608** for certificate bundle distribution. In response to the request, the certificate transmitting engine **612** may transmit the certificate bundle to one or more network entities **608** including at least the particular network entity **608** associated with the request.

(99) In one example, the certificate transmitting engine **612** may determine that a new certificate bundle has been stored in the certificate repository **604** based on a file pointer in a configuration file. The file pointer may point to a location in the certificate repository **604** where a new certificate bundle is stored. The configuration file and/or the file pointer in the configuration file may be updated when a new certificate bundle is stored in the certificate repository **604**. For example, the file pointer may point to a different location in the certificate repository **604** when a new certificate bundle is stored in the certificate repository **604**. The certificate transmitting engine **612** may periodically poll the configuration file for an updated file pointer. The file pointer may be

compared to an earlier file pointer. A difference between a current file pointer and an earlier file pointer may indicate that a new certificate bundle has been added to the certificate repository **604**. A trigger condition indicating that a new certificate bundle is available in the certificate repository **604** may be based at least in part on a current file pointer differing from an earlier file pointer.

(100) In one example, the certificate transmitting engine **612** may determine that a new certificate bundle has been stored in the certificate repository **604** based on a timestamp, such as a timestamp of a configuration file. The certificate transmitting engine **612** may periodically poll the certificate repository **604** and/or the configuration file for a new timestamp corresponding to a new certificate bundle having been stored in the certificate repository **604**. The new timestamp may be compared to an earlier timestamp. A difference between the new timestamp and the earlier timestamp may indicate that a new certificate bundle has been added to the certificate repository **604** subsequent to the time of the earlier timestamp. A trigger condition indicating that a new certificate bundle is available in the certificate repository **604** may be based at least in part on the new timestamp differing from, such as being subsequent to, the earlier timestamp.

(101) The certificate repository **604** may include a plurality of certificate bundles. In one example, the certificate repository **604** may include a new certificate bundle **614** and an earlier certificate bundle **616**. Each certificate bundle may include one or more CA certificates. The CA certificates may include root CA certificates and/or intermediate CA certificates.

(102) As used herein, the term “new certificate bundle” refers to a certificate bundle that, with respect to at least one network entity **608**, has yet to be distributed to the at least one network entity. A new certificate bundle **614** may include a set of one or more CA certificates. The set of one or more CA certificates in a new certificate bundle may include at least one CA certificate that has yet to be included in a certificate bundle distributed to the at least one network entity **608**.

(103) As used herein, the term “earlier certificate bundle” refers to a certificate bundle that, with respect to at least one network entity **608**, has previously been distributed to the at least one network entity. An earlier certificate bundle **616** may include a set of one or more CA certificates. The set of one or more CA certificates in an earlier certificate bundle **616** may include at least one CA certificate included in a certificate bundle that was previously distributed to the at least one network entity **608**.

(104) In addition, or in the alternative, to a new certificate bundle **614** and an earlier certificate bundle **616**, the certificate repository **604** may include one or more certificate bundles that include a set of CA certificates for use in connection with a particular set of network entities. For example, the certificate repository **604** may include a first set of one or more certificate bundles that include a first set of CA certificates for use by a first set of network entities, and a second set of one or more certificate bundles that include a second set of CA certificates for use by a second set of network entities. In one example, the first set of one or more certificate bundles may include a first new certificate bundle **614** and a first earlier certificate bundle **616**. Additionally, or in the alternative, the second set of one or more certificate bundles may include a second new certificate bundle and a second earlier certificate bundle.

(105) The certificate repository **604** and/or the distribution schedule repository **606** may include any type of storage unit and/or device (e.g., a file system, database, collection of tables, or any other storage mechanism) for storing data. Further, the certificate repository **604** and/or the distribution schedule repository **606** may include multiple different storage units and/or devices. The multiple different storage units and/or devices may or may not be of the same type or located at the same physical site. The certificate repository **604** and the distribution schedule repository **606** may share one or more storage units. Additionally, or in the alternative, the certificate repository **604** may include one or more storage units that differ from one or more storage units of the distribution schedule repository **606**. Further, the certificate repository **604** and/or the distribution schedule repository **606** may be implemented or executed on the same computing system as the distribution service **602** and/or the plurality of network entities **608**. Additionally, or in the

alternative, the certificate repository **604** and/or the distribution schedule repository **606** may be implemented or executed on a computing system separate from the distribution service **602** and/or the plurality of network entities **608**. The certificate repository **604** and/or the distribution schedule repository **606** may be communicatively coupled to the distribution service **602** and/or the plurality of network entities **608** via a direct connection or via a network.

(106) In one example, the distribution schedule generator **610** may determine, for each particular network entity **608** of the plurality of network entities **608** associated with the virtual cloud network **601**, a particular certificate distribution group from among a plurality of certificate distribution groups based on a set of one or more partitioning criteria. In one example, the distribution schedule generator **610** may additionally determine, for a set network entities **608** associated with a particular certificate distribution group, a certificate distribution subgroup form among a plurality of certificate distribution subgroups. A distribution schedule for transmitting a certificate bundle to the plurality of network entities **608** may include a series of release phases that respectively correspond to a particular certificate distribution group and/or a particular certificate distribution subgroup.

(107) In one example, the certificate transmitting engine **612** may select a certificate distribution group, from among the plurality of certificate distribution groups, for distribution of a certificate bundle. The certificate distribution group may be selected based on a distribution schedule from the distribution schedule repository **606**. In one example, upon having selected a certificate distribution group, the certificate transmitting engine **612** may select a certificate distribution subgroup of the certificate distribution group, from among a plurality of certificate distribution subgroups, for distribution of the certificate bundle. The certificate distribution subgroup may be selected based on a distribution schedule from the distribution schedule repository **606**.

(108) In one example, the certificate transmitting engine **612** may transmit a certificate bundle to the certificate distribution groups and/or to the certificate distribution subgroups in a series of release phases. The certificate distribution group and/or the certificate distribution subgroup selected by the certificate transmitting engine **612** may correspond to a current release phase, for example, of a series of release phases for distributing the certificate bundle to the plurality of network entities **608**. In one example, with respect to a first release phase, the certificate transmitting engine **612** may select a first certificate distribution group and transmit the certificate bundle to a first subset of network entities in the first certificate distribution group. In one example, with respect to the first certificate distribution group, the certificate transmitting engine **612** may select a first certificate distribution subgroup corresponding to a first release phase, and may transmit the certificate bundle to a first subgroup of network entities in the first certificate subgroup. In subsequent release phases, the certificate transmitting engine **612** may proceed through the plurality of certificate distribution subgroups of the first certificate distribution group prior to proceeding to a next certificate distribution group. Additionally, or in the alternative, with respect to a subsequent release phase, the certificate transmitting engine **612** may select a second certificate distribution group and may transmit the certificate bundle to a second subset of network entities in the second certificate distribution group.

(109) In one example, the distribution service **602** may detect a trigger condition to distribute a certificate bundle. The trigger condition may include a command, such as in an executable file associated with a certificate distribution process, to distribute a certificate bundle to one or more the network entities **608**. Additionally, or in the alternative, the trigger condition may include the distribution service having determined a presence of a new certificate bundle **614** in the certificate repository **604**. Additionally, or in the alternative, the trigger condition may include a request from a network entity **608** for the distribution service **602** to distribute a certificate bundle to the network entity **608**. In one example, the distribution schedule generator **610** may generate a distribution schedule in response to the distribution service **602** detecting the trigger condition. Additionally, or in the alternative, the distribution schedule generator **610** may generate a distribution schedule prior

to the distribution service **602** detecting the trigger condition. In one example, the certificate transmitting engine **612** may transmit a certificate bundle to one or more network entities in response to the distribution service **602** detecting the trigger condition.

(110) As shown in FIG. **6A**, a network entity **608** may include a provisioning module **618** and a storage medium **620**. The provisioning module **618** may perform an initial provisioning of the network entity **608**. Additionally, or in the alternative, the provisioning module **618** may perform periodic updates to the network entity **608**. An initial provisioning and/or a periodic update of a network entity **608** may include obtaining a certificate bundle and installing the certificate bundle in the storage medium **620** associated with the network entity **608**. The provisioning module **618** may request a certificate bundle from the distribution service **602**. The request for a certificate bundle may be associated with an initial provisioning of the network entity **608** or a periodic update to the network entity **608**. The distribution service **602** may distribute a certificate bundle to the provisioning module **618** responsive to a request from the provisioning module **618**. Additionally, or in the alternative, the distribution service may push certificate bundles to a provisioning module **618**, for example, without receiving a request from the provisioning module **618**.

(111) The storage medium **620** may include any type of storage unit and/or device (e.g., a file system, database, collection of tables, or any other storage mechanism) for storing data. Further, the storage medium **620** may include multiple different storage units and/or devices. The multiple different storage units and/or devices may or may not be of the same type or located at the same physical site. The storage medium **620** may include one or more storage units. Further, the storage medium **620** may be implemented or executed on the same computing system as the provisioning module **618**. Additionally, or in the alternative, the storage medium **620** may be implemented or executed on a computing system separate from the provisioning module **618**. The storage medium **620** may be communicatively coupled to the provisioning module **618** via a direct connection or via a network.

B. Partitioning Criteria for Certificate Distribution Groups

(112) In one example, the partitioning criteria for determining certificate distribution groups corresponding to network entities may include a fault domain of the particular network entity. For example, the partitioning criteria may include limiting each particular certificate distribution group, of the plurality of certificate distribution groups, to a specific fault domain of the computer network. As used herein, the term “fault domain” refers to a portion of a computer network, such as a virtual cloud network, that shares a common single point of failure. The virtual cloud network **601** may include a plurality of fault domains that respectively represent a single point of failure. Each particular certificate distribution group may include a particular subset of network entities **608** that are associated with a particular single point of failure corresponding to a particular fault domain. A fault domain corresponding to a particular network entity **608** may be determined based on metadata associated with the particular network entity **608**. The metadata may include a fault domain attribute that identifies the particular fault domain corresponding to the particular network entity **608**.

(113) In one example, the partitioning criteria for determining certificate distribution groups corresponding to network entities may include a resource group of the particular network entity. For example, the partitioning criteria may include limiting each particular certificate distribution group, of the plurality of certificate distribution groups, to a specific resource group of the computer network. As used herein, the term “resource group” refers to a subset of network entities of a computer network, such as a virtual cloud network, that represent an attribute, a functionality, or a purpose of the computer network. The virtual cloud network **601** may include a plurality of resource groups that respectively represent at least one of: an attribute, a functionality, or a purpose, of the virtual cloud network **601**. Each particular certificate distribution group may include a particular subset of network entities **608** that are associated with a particular resource group. A

resource group corresponding to a particular network entity **608** may be determined based on metadata associated with the particular network entity **608**. The metadata may include a resource group attribute that identifies the particular resource group corresponding to the particular network entity **608**. A resource group may be distributed across a plurality of fault domains. Additionally, or in the alternative, a fault domain may include a plurality of resource groups.

(114) In one example, a plurality of resource groups associated with a computer network, such as virtual cloud network **601**, may include at least one of: a security services resource group, an identity and access management resource group, a data processing resource group, a database resource group, a testing and staging environment resource group, a production resource group, a development environment resource group, a virtual machine resource group, a high-performance computing resource group, an API gateway and microservices resource group, a web applications resource group, a tenant resource group, a content delivery network resource group, an internet-of-things (IoT) device management resource group, a data analytics resource group, or a backup and disaster recovery resource group.

(115) A security services resource group may include network entities related to security, such as firewalls, intrusion detection systems, or security monitoring tools. An identity and access management resource group may include network entities related to managing user identities, authentication, authorization, and access control. A data processing resource group may include network entities related to data processing, such as data ingestion, data transformation, or data generation. A database resource group may include network entities related to data storage, data maintenance, database clusters, database replication, load balancers, or failover mechanisms. A testing and staging environment resource group may include network entities related to testing, quality assurance, or staging. A production resource group may include network entities related to customer-facing services, applications, or resources. Such customer-facing services, applications, or resources may include virtual machines, containers, or serverless instances, application servers, database servers, load balancers, auto-scaling services, caching services, metrics services, deployment services, provisioning services, discovery services, or external integration services. A development environment resource group may include network entities related to software development, such as network entities related to development servers, version control repositories, or integration tools. A high-performance computing resource group may include network entities related to specialized virtual machines, high-speed interconnects, or parallel processing frameworks. An API gateway and microservices resource group may include network entities related to API gateways, API-driven applications, microservices, or serverless functions. A web applications resource group may include network entities related to web application, or resources that collectively form a web application, such as virtual machines, load balancers, or databases. A tenant resource group may include network entities related to resources that are dedicated to a specific tenant or a specific set of tenants, for example, within a multi-tenant environment. A content delivery network resource group may include network entities related to content distribution and delivery, content delivery network endpoints, origin servers, or caching. An internet-of-things (IoT) device management resource group may include network entities related to managing IoT systems, monitoring, control, device registries, message queues, or data pipelines. A data analytics resource group may include network entities related to big data analytics tasks, such as data lakes, clusters, or analytical databases. A backup and disaster recovery resource group may include network entities related to backup storage, disaster recovery sites, or replication mechanisms.

(116) In one example, one or more resource groups may be combined or divided to form another resource group. Additionally, or in the alternative, a resource group may include a plurality of resource subgroups. Additionally, or in the alternative, a resource subgroup may be configured as a resource group.

(117) In one example, a security services resource group may include a plurality of resource

subgroups. The plurality of resource subgroups in a security services resource group may include one or more of: a user and group management resource subgroup, an authentication services resource subgroup, an authorization services resource subgroup, an identity federation resource subgroup, a security token services resource subgroup, a directory services resource subgroup, an access policies and rules resource subgroup, an audit and logging resource subgroup, a certificate authority resource subgroup, a key management services resource subgroup, a user provisioning and deprovisioning resource subgroup, or a compliance and governance tools resource subgroup.

(118) A user and group management resource subgroup may include network entities related to resources for managing user accounts, groups, roles, or permissions, and/or network entities related to user directories, identity providers, or access control lists. An authentication services resource subgroup may include network entities related to user authentication, authentication services, or authentication servers. An authorization services resource subgroup may include network entities related to enforcing authorization policies, policy engines, access controls, access control systems, or access control mechanisms. An identity federation resource subgroup resource subgroup may include network entities related to resources for establishing trust relationships between different identity domains. A security token services resource subgroup resource subgroup may include network entities related to services that issue security tokens for authentication or authorization purposes. A directory services resource subgroup resource subgroup may include network entities related to directory services that store and manage user and group information. An access policies and rules resource subgroup resource subgroup may include network entities related to configuration or management of access policies or rules. An audit and logging resource subgroup may include network entities related to resources for tracking, monitoring, reporting, or auditing identity and access management activities. A certificate authority resource subgroup may include network entities related to management or maintenance of a certificate authority infrastructure, such as network entities related to issuing or managing digital certificates used for secure communication, authentication, or encryption. A key management services resource subgroup may include network entities related to issuance or management of cryptographic keys used for secure communication, authentication, or encryption. A user provisioning and deprovisioning resource subgroup may include network entities related to creation, modification, or removal of user accounts and/or access permissions associated with user accounts. A compliance and governance tools resource subgroup may include network entities related to identity and access management policies.

(119) In one example, the partitioning criteria for determining certificate distribution groups may include an availability domain of the particular network entity. For example, the partitioning criteria may include limiting each particular certificate distribution group, of the plurality of certificate distribution groups, to a specific availability domain of the computer network. As used herein, the term “availability domain” refers to a physically separate portion of a computer network, such as a virtual cloud network, with respect to one or more failure modes. An availability domain may include a plurality of fault domains. Additionally, or in the alternative, an availability domain may include a plurality of resource groups.

(120) In one example, a set of network entities may be assigned to one of a plurality of certificate distribution classes based on an availability domain corresponding to the particular network entity. Each certificate distribution class may include a plurality of certificate distribution groups. For example, a first certificate distribution class corresponding to a first availability domain may include a first set of certificate distribution groups, and a second certificate distribution class corresponding to a second availability domain may include a second set of certificate distribution groups. The network entities in the first certificate distribution class and/or in the first set of certificate distribution groups may be located in the first availability domain. The network entities in the second certificate distribution class and/or in the second set of certificate distribution groups may be located in the second availability domain.

(121) The virtual cloud network **601** may include a plurality of availability domains that respectively represent a physically separate portion of the virtual cloud network **601** with respect to one or more failure modes. Each particular certificate distribution group may include a particular subset of network entities **608** that are associated with a particular availability domain. An availability domain corresponding to a particular network entity **608** may be determined based on metadata associated with the particular network entity **608**. The metadata may include an availability domain attribute that identifies the particular availability domain corresponding to the particular network entity **608**.

C. Example Certificate Distribution Groups

(122) Referring again to FIG. **6A**, a virtual cloud network **601** may include a plurality of fault domains **622**. Each fault domain **622** may include a set of network entities **608**. Additionally, or in the alternative, the virtual cloud network **601** may include a plurality of resource groups **624**. Each resource group may include a set of network entities **608**. A fault domain **622** may include a plurality of resource groups **624**. Additionally, or in the alternative, a resource group **624** may be distributed across a plurality of fault domains **622**. The plurality of fault domains **622** and/or the plurality of resource groups may correspond to a particular availability domain **626**. A network entity may include a plurality of availability domains.

(123) The network entities **608** of the virtual cloud network **601** may be partitioned into a plurality of certificate distribution groups **628**. In one example, as shown in FIG. **6A**, a first certificate distribution group **628a** may include a set of network entities **608** corresponding to fault domain **622a**. Additionally, or in the alternative, the first certificate distribution group **628a** may include resource group **624a** and resource group **624n**. Network entity **608a** and network entity **608b** may correspond to fault domain **622a** and/or to resource group **624a**. Network entity **608n** and network entity **608z** may correspond to fault domain **622** and/or to resource group **624n**. The first certificate distribution group **628a** may include network entity **608a**, network entity **608b**, network entity **608n**, and network entity **608z**.

(124) Additionally, or in the alternative, as further shown in FIG. **6A**, a second certificate distribution group **628b** may include a portion of resource group **624c**. Resource group **624c** may be distributed across fault domain **622c** and fault domain **622e**. The second certificate distribution group **628b** may include the portion of resource group **624c** corresponding to fault domain **622c**. Network entity **608c** and network entity **608d** may correspond to fault domain **622c** and/or to resource group **624c**. The second certificate distribution group **628b** may include network entity **608c**, and network entity **608d**. In one example, a third certificate distribution group **628c** may include the portion of resource group **624c** corresponding to fault domain **622e**. Network entity **608e** and network entity **608f** may correspond to fault domain **622e** and/or to resource group **624c**. The third certificate distribution group **628c** may include network entity **608e**, and network entity **608f**.

D. Partitioning Criteria for Certificate Distribution Subgroups

(125) In one example, the distribution schedule generator **610** may determine, for each particular network entity **608** associated with a particular certificate distribution group **628**, a particular certificate distribution subgroup, from among a plurality of certificate distribution subgroups, corresponding to the particular network entity. The particular certificate distribution subgroup may be determined based on a network address of the particular network entity **608**. Each particular certificate distribution subgroup may include a corresponding subgroup of network entities **608** from the set of network entities in the certificate distribution group.

(126) The distribution schedule generator **610** may determine, for each particular network entity **608** of a particular certificate distribution group, a particular certificate distribution subgroup from among a plurality of certificate distribution subgroups, at least by applying a randomization function to the network address of the particular network entity. The network addresses of the network entities may be determined by using an API to query a network management system

associated with the virtual cloud network **601**. Additionally, or in the alternative, the network addresses of the network entities may be determined using a DNS lookup function or a DNS look up table. Additionally, or in the alternative, the network addresses of the network entities may be determined by referencing a host configuration log. For example, network entities may be automatically assigned an IP address when they are provisioned or booted-up. The IP addresses may be assigned according to a dynamic host configuration protocol. Additionally, or in the alternative, the network addresses of the network entities may be determined by querying metadata associated with the network addresses.

(127) A product of the randomization function may be used to determine a particular certificate distribution subgroup of the plurality of certificate distribution subgroups for the particular network entity **608**. In one example, the randomization function may include a hash function and a modulo function. The hash function may generate a hash value of the network address for a particular network entity **608**. The modulo function may be executed on the hash value of the network address. The modulo function may output a remainder, and the certificate distribution subgroup may be selected based at least in part on the remainder corresponding to the particular network entity **608**. As used herein, the term “modulo function” refers to a mathematical operation that computes a remainder after division of a dividend by a divisor. A modulo function may be represented as “ $A \bmod B$,” where “A” is the dividend and “B” is the divisor. The product of the modulo function is the remainder obtained when “A” is divided by “B.” The remainder represents what is left over after the division of the dividend by the divisor. As an example, for $15 \bmod 7=1$, where dividing 15 by 7 gives a quotient of 2 and a remainder of 1. As another example, $27 \bmod 5=2$, where dividing 27 by 5 gives a quotient of 5 and a remainder of 2. In one example, the remainder may be rounded to the nearest whole number.

(128) In one example, a quantity of certificate distribution subgroups in the plurality of certificate distribution subgroups may correspond to a modulus base of the modulo function. As used herein, the term “modulus base” refers to the divisor “B” in the modulo function. The modulus base represents the number by which the dividend, “A,” is divided. In one example, each particular certificate distribution subgroup of the plurality of certificate distribution subgroups may correspond to a particular remainder from among the modulus base. Additionally, or in the alternative, each particular certificate distribution subgroup may include a particular subgroup of network entities corresponding to one or more remainder values from among the modulus base. In one example, each particular certificate distribution subgroup may correspond to a particular remainder value. For example, a first certificate distribution subgroup may include network entities corresponding to a remainder value of 1, a second certificate distribution subgroup may include network entities corresponding to a remainder value of 2, and an Nth certificate distribution subgroup may include network entities corresponding to a remainder value of N, where N is the modulus base. Additionally, or in the alternative, each certificate distribution subgroup may correspond to a plurality of remainder values obtainable from the modulo function. For example, a first certificate distribution subgroup may include network entities corresponding to a remainder value within the range of 1 to N/x , a second certificate distribution subgroup may include network entities corresponding to a remainder value within the range of $1+N/x$ to $2(N/x)$, and an Nth certificate distribution subgroup may include network entities corresponding to a remainder value of $x-1$ to N, where “N” is the modulus base and “x” is the number of certificate distribution subgroups.

(129) In one example, a randomization function may provide a random distribution of network entities across the plurality of certificate distribution subgroups. For example, the hash function may generate a unique, or random hash value from the network address corresponding to a network entity. Additionally, or in the alternative, executing of the modulo function on the hash values corresponding to a plurality of network entities may result in an even distribution of remainder values resulting from the modulo function. The allocation of the network entities to a certificate

distribution subgroup corresponding to the remainder resulting from the modulo function may result in an even distribution of network entities across the certificate distribution subgroups.

(130) In one example, the modulus base may be selected based on a desired quantity of network entities in a certificate distribution subgroup. Additionally, or in the alternative, the modulus base may be selected based on a desired percentage of the total quantity of network entities, in the set of network entities, in a certificate distribution subgroup. In one example, the quantity of network entities in a certificate distribution subgroup may differ as between respective certificate distribution subgroups. In one example, the modulus base may be selected based at least in part on an acceptable level of variance in the quantity of network entities in a certificate distribution subgroup. In one example, the modulus base may be 100. In one example, the modulus base may be 1,000.

(131) In one example, a quantity of remainders corresponding to particular certificate distribution subgroups may be equivalent as between at least some of the certificate distribution subgroups. For example, the plurality of certificate distribution subgroups may be spread evenly across the modulus base such that each certificate distribution subgroup includes an equivalent quantity of remainders. Alternatively, a quantity of remainders corresponding to a particular certificate distribution subgroup may differ as between at least some of certificate distribution subgroups. For example, the plurality of certificate distribution subgroups may be spread disproportionately across the modulus base such that at least some of the certificate distribution subgroup includes a different quantity of remainders relative to one or more of the other certificate distribution subgroups. In one example, a first certificate distribution subgroup may include a first subgroup of network entities that correspond to a first set of one or more remainder values from among the modulus base, and wherein a second certificate distribution subgroup may include a second subgroup of network entities that correspond to a second set of one or more remainder values from among the modulus base. A first quantity of remainders in the first set of one or more remainder values may differ from, and/or may be less than, a second quantity of remainder values in the second set of one or more remainder values. In one example, a distribution schedule may include a series of certificate distribution subgroups that respectively include an increasing quantity of remainder values from among the modulus base. As a result of the increasing quantity of remainder values from among the modulus base, the particular certificate distribution subgroups of the series of certificate distribution subgroups may include an increasing quantity of network entities. In one example, the quantity of remainder values and/or the quantity of network entities in the respective certificate distribution subgroups of the series of certificate distribution subgroups may correspond to an exponential function and/or a recursive function. An exponential function may include at least one of: a base exponential function (e.g., $f(x)=a^{\circ}x$, where “a” is a constant base greater than zero and not equal to 1), a natural exponential function (e.g., $f(x)=e^{\circ}x$, where “e” is the base of the natural logarithm), a logistic function, a Gompertz function, a Weibull function. A recursive function may include a function that can be defined in terms of itself. A recursive function may include at least one of: a Fibonacci sequence, a factorial function, an Ackermann function, or a Collatz conjecture. In one example, a recursive function may include an exponential function, such as $f(x)=a*f(x-1)$, where “a” is a constant base multiplied by the previous term, $f(x-1)$.

(132) The hash function utilized in the randomization function may include a cryptographic hash function or a non-cryptographic hash function. The hash function utilized in the randomization function may be selected based on one or more of the following properties: determinism, irreversibility, or avalanche effect. Determinism refers to the same input producing the same hash value. Irreversibility refers to difficulty in deriving an original input from the hash value. Avalanche effect refers to small changes in an input resulting in significant changes in the hash value. An example hash function may include a cryptographic hash function, such as an SHA-256 hash function, an Argon2 hash function, a Message Digest Algorithm 5 hash function, a RIPEMED

hash function, a Whirlpool hash function, a Tiger hash function, a SipHash function, a BLAKE2 hash function, or an SHA3 (e.g., SHA3-256) hash function. By way of illustration, an SHA-256 or an SHA3-256 hash function may transform an input into a 256-bit output. Additionally, or in the alternative, an example hash function may include a non-cryptographic hash function, such as a MurmurHash hash function, a CityHash hash function, a Jenkins hash function, an xxHash hash function, a Fowler-Noll-Vo hash function, or a CRC32 hash function.

D. Example Distribution Schedules

(133) Referring now to FIGS. 6B and 6C, example distribution schedules **630** are described. An example distribution schedule **630** for releasing a certificate bundle for distribution to a plurality of certificate distribution groups is described with reference to FIG. 6B. An example distribution schedule **630** for releasing the certificate bundle for distribution to a plurality of certificate distribution subgroups is described with reference to FIG. 6C.

(134) As shown in FIG. 6B, a first distribution schedule **630a** may include a plurality of certificate distribution groups **632**. Each certificate distribution group **632** may include a subset of network entities **636**. Each certificate distribution group **632** may correspond to a release phase **634** of the distribution schedule **630a**. The subset of network entities **636** in a certificate distribution group **632** may correspond to a fault domain **638** and/or to a resource group **640**.

(135) In one example, a certificate distribution schedule **630a** may include a series of release phases **634** for distributing a certificate bundle to a plurality of certificate distribution groups **632** that respectively include a subset of network entities **636** corresponding to a particular resource group **640** that is distributed across a plurality of fault domains **638**. According to the distribution schedule **630a**, the certificate bundle may be distributed to network entities **636** corresponding a particular resource group **640** distributed across a plurality of fault domains **638**, with a separate certificate distribution group corresponding to each of the fault domains **638**. After the certificate bundle has been distributed to the network entities **636** corresponding to the particular resource group **640** at each of a plurality of fault domains **638**, the certificate bundle may be distributed to network entities **636** associated with another particular resource group **640**, which resource group **640** may also be distributed across a plurality of fault domains **638**.

(136) As shown in FIG. 6B, the certificate distribution schedule **630a** may include a first certificate distribution group **632a** corresponding to a first release phase **634a**. The first certificate distribution group **632a** may include a first subset of network entities **636a** that correspond to a first fault domain **638a** and a first resource group **640a**. Additionally, or in the alternative, the certificate distribution schedule **630a** may include a second certificate distribution group **632b** corresponding to a second release phase **634b**. The second certificate distribution group **632b** may include a second subset of network entities **636b** that correspond to the first resource group **640a** and a second fault domain **638b**. Additionally, or in the alternative, the certificate distribution schedule **630a** may include a third certificate distribution group **632c** corresponding to a third release phase **634c**. The third certificate distribution group **632c** may include a third subset of network entities **636c** that correspond to the first fault domain **638a** and the second resource group **640b**.

Additionally, or in the alternative, the certificate distribution schedule **630a** may include a fourth certificate distribution group **632d** corresponding to a fourth release phase **634d**. The fourth certificate distribution group **632d** may include a fourth subset of network entities **636d** that correspond to the second fault domain **638b** and the second resource group **640b**.

(137) Additionally, or in the alternative, a certificate distribution schedule **630a** may include a series of release phases **634** for distributing a certificate bundle to a plurality of certificate distribution groups **632** that respectively include a subset of network entities **636** corresponding to a particular fault domain **638** that includes a plurality of resource groups **640**. According to the distribution schedule **630a**, the certificate bundle may be distributed to network entities **636** corresponding a particular fault domains **638**, with a separate certificate distribution group corresponding to each resource group **640** of the fault domains **638**. After the certificate bundle has

been distributed to the network entities **636** corresponding to the particular resource group **640** in the particular fault domain **638**, the certificate bundle may be distributed to network entities **636** corresponding to another particular resource group **640** in the particular fault domain **638**. After the certificate bundle has been distributed to network entities **636** corresponding to each resource group **640** in the particular fault domain **638**, the certificate bundle may be distributed to another set of certificate distribution groups corresponding to another fault domain **638**.

(138) In one example, as shown in FIG. **6B**, a certificate distribution schedule **630a** may include one or more release phases **634**, such as release phase **634e**, corresponding to one or more pre-distribution groups **642**. The one or more pre-distribution groups **642** may include a subset of network entities **636**, such as network entities **636e**, selected for pre-distribution. A subset of network entities **636** may be selected for pre-distribution based on one or more pre-distribution criteria. The one or more pre-distribution criteria may be intended to identify network entities **636** to receive the certificate bundle in the pre-distribution, for example, to identify potential issues with distribution of the certificate bundle and/or to assess compatibility of the certificate bundle with respect to various features of the virtual cloud network. The pre-distribution may reduce a risk of widespread issues in the event of a problem with the certificate bundle. In one example, the network entities **636** in a pre-distribution group may include network entities that are not associated with customer-facing aspects of the virtual cloud network. Such network entities are referred to herein as “internal network entities.” In one example, the network entities **636** in a pre-distribution group may include one or more network entities associated with one or more particular resource groups **640**. The pre-distribution may include testing the distribution of the certificate bundle with respect to the one or more particular resource groups **640** to identify potential issues with distribution of the certificate bundle to the one or more particular resource groups **640** and/or to assess compatibility of the certificate bundle with respect to the one or more particular resource groups **640**. In one example, the network entities **636** in a pre-distribution group may include one or more network entities associated with one or more particular fault domains **638**. The pre-distribution may include testing the distribution of the certificate bundle with respect to the one or more particular fault domains **638** to identify potential issues with distribution of the certificate bundle to the one or more particular fault domains **638** and/or to assess compatibility of the certificate bundle with respect to the one or more particular fault domains **638**. In one example, a certificate bundle may be tested in a pre-distribution to at least one network entity **636** in each resource group **640** and/or in each fault domain **638** prior to distributing the certificate bundle to certificate distribution groups **635** associated with the particular resource group **640** and/or fault domain **638**. Additionally, or in the alternative, a certificate bundle may be tested in a pre-distribution to at least one internal network entity **636** prior to distributing the certificate bundle to certificate distribution groups **635** associated with customer-facing aspects of the virtual cloud network.

(139) In one example, as shown in FIG. **6B**, a certificate distribution schedule **630a** may include one or more release phases **634**, such as release phase **634n**, corresponding to one or more aft-distribution groups **644**. The one or more aft-distribution groups **644** may include a subset of network entities **636**, such as network entities **636n**, that are not included in the set of certificate distribution groups, for example, due to not fitting into the one or more partitioning criteria corresponding to the certificate distribution groups **632**. Additionally, or in the alternative, the one or more network entities **636n** in an aft-distribution group may have particular characteristics, requirements, or usage patterns. In one example, the one or more network entities **636n** in an aft-distribution group may not fit into the certificate distribution groups **632** as a result of the particular characteristics, requirements, or usage patterns associated with the one or more network entities **636n**.

(140) In one example, one or more network entities **636** may be excluded from the plurality of certificate distribution group, for example, even though the one or more network entities **636** may

otherwise meet the one or more partitioning criteria corresponding to the certificate distribution groups **632**. Network entities **636** that are excluded from a certificate distribution group **632** may be assigned to an aft-distribution group **644**. One or more network entities **636** may be excluded from a certificate distribution group **632** based on particular characteristics, requirements, or usage patterns associated with the one or more network entities **636**. Additionally, or in the alternative, one or more network entities **636** may be excluded from a certificate distribution group **632** to avoid potential issues, conflicts, or incompatibility. The one or more network entities **636** may be excluded from a certificate distribution group **632** to meet requirements of other network entities **636** in the certificate distribution group and/or to avoid potential impacts on the overall group of network entities **608** in a certificate distribution group **632** that may be attributable to the one or more particular network entities **636**. Additionally, or in the alternative, one or more network entities **636** that are of particular importance may be excluded from a certificate distribution group **632**, for example, so that the certificate bundle may be distributed to those particular important network entities **636** in a separate release phase **634** and/or so that those particular important network entities **636** are not impacted by potential issues related to distributing the certificate bundle to the other network entities **636** in the certificate distribution group **632**.

(141) In one example, one or more network entities **636** may be assigned to an aft-distribution group **644** based on a magnitude of an impact that an issue with respect to those one or more network entities **636** may have on the virtual cloud network and/or customers of the virtual cloud network. For example, one or more network entities **636** associated with critical resources or services may be assigned to an aft-distribution group **644**. Additionally, or in the alternative, one or more network entities **636** associated with a large group of customers, and/or a particularly important customer, may be assigned to an aft-distribution group **644**. **636n**

(142) Referring now to FIG. **6C**, a second distribution schedule **630b** may include a plurality of certificate distribution subgroups **646**. Network entities of a particular certificate distribution group **632** (FIG. **6B**) may be assigned to one of a plurality of certificate distribution subgroups **646**. Each certificate distribution subgroup **646** may include a subgroup of network entities **636**, from a particular certificate distribution group **632** (FIG. **6B**), that are assigned to a particular certificate distribution subgroup **646** based on a network address of the particular network entity **636**. Each certificate distribution subgroup **646** may correspond to a release phase **634** of the distribution schedule **630a**. The subgroup of network entities **636** in a certificate distribution subgroup **646** may correspond to a fault domain **638** and/or to a resource group **640** of the certificate distribution group **632** (FIG. **6B**) from which the subgroup of network entities **636** are assigned.

(143) As shown in FIG. **6C**, a second distribution schedule **630b** may include a plurality of certificate distribution subgroups **646**. The plurality of certificate distribution subgroups **646** may correspond to a certificate distribution group **632** in the first distribution schedule **630a** (FIG. **6B**). The second distribution schedule **630b** may include a first certificate distribution subgroup **646x**, a second certificate distribution subgroup **646y**, and/or a third certificate distribution subgroup **646z**. The first certificate distribution subgroup **646x** may include a first subgroup of network entities **648x** corresponding to release phase **634x**. The first subgroup of network entities **648x** may be assigned to the first certificate distribution subgroup **646x** based on the network addresses **649** of the network entities. The second certificate distribution subgroup **646y** may include a second subgroup of network entities **648y** corresponding to release phase **634y**. The second subgroup of network entities **648y** may be assigned to the second certificate distribution subgroup **646y** based on the network addresses **649** of the network entities. The third certificate distribution subgroup **646z** may include a third subgroup of network entities **648z** corresponding to release phase **634z**. The third subgroup of network entities **648z** may be assigned to the third certificate distribution subgroup **646z** based on the network addresses **649** of the network entities.

E. Further Example Virtual Cloud Network Architecture

(144) Referring now to FIG. **6D**, the system **600** is further described with reference to an example

virtual cloud network **650**. The virtual cloud network **650** may include the virtual cloud network **601** described with reference to FIG. **6A**. As shown in FIG. **6D**, the virtual cloud network **650** may include a substrate network **652**, an overlay network **654**, and a network interface **656** that provides communication between the substrate network **652** and the overlay network **654**. The system may include a plurality of network entities **658** located throughout the virtual cloud network. The plurality of network entities **658** may correspond to the plurality of network entities **608** described with reference to FIG. **6A**. A network entity **658** may reside on the substrate network **652**, the overlay network **654**, or the network interface **656**. A network entity **658** may be implemented in hardware and/or software in association with the virtual cloud network, such as a node, a host, an agent, a service, a component, an endpoint, or other element. The plurality of network entities **658** may include one or more substrate entities **660**, one or more interface entities **662**, and/or one or more overlay entities **664**.

(145) As used herein, the term “substrate entity” refers to a network entity **658** implemented in a substrate network **652**. As used herein, the term “substrate network” refers to a physical network infrastructure. The substrate network generally provides a foundation of a virtual cloud network. The substrate network **652** may include physical network devices, such as routers, switches, network links, and other networking components. The substrate network **652** may generally provide the basic connectivity and transport capabilities necessary for data transmission within and between data centers.

(146) The one or more substrate entities **660** may include substrate hosts, routers, firewall appliances, load balancers, storage devices, and/or substrate services. A substrate host may include an endpoint within the substrate network **652**, such as a bare metal host, a virtual machine, a container, or a physical server. A substrate service may include a service executing or executable on a substrate entity, such as a firmware service, a network connectivity service, an addressing service, a name resolution service, a security service, a network monitoring service, a load balancing service, and/or a storage service. A firmware service may be associated with functionality or management of network infrastructure components or services, such as network devices, boot-up or initialization process, hardware controls, feature enablement, updates, hardware abstraction, network configuration, and/or network management. In one example, a substrate entity **660** may include a combination of hardware and software. In one example, the one or more substrate entities **660** may include one or more substrate hosts, and/or one or more substrate services. In one example, a substrate host may include a bare metal host. In one example, a substrate service may include a firmware service. The substrate entities **660** may communicate with one another, and/or with other network entities **658**, using logical network addresses assigned within the overlay network.

(147) As used herein, the term “network interface” refers to a communication interface between a substrate network and an overlay network, such as a network interface card, a smartNIC, or the like. A network interface **656** may include one or more interface entities **662**, such as a node on the network interface **656**, or an interface service executing or executable on the network interface **656**. A node on the network interface **656** may include a programmable hardware component, a memory component, or a gateway component. In one example, a network interface **656** may include a network interface card, such as a smartNIC. Additionally, or in the alternative, a network interface **656** may include a node or an endpoint on a network interface card or smartNIC.

(148) A gateway component may provide connectivity between the substrate network **652** and the network interface **656**, and/or between the network interface **656** and the overlay network **654**. For example, a gateway component may enable communication between overlay entities **664** and substrate entities **660**. Additionally, or in the alternative, a gateway component may provide connectivity between the overlay network **654** and external networks, such as the internet or other networks outside the overlay network. For example, an overlay gateway may enable communication between overlay entities **664** and external endpoints.

(149) As used herein, the term “overlay network” refers to a virtual network built on a substrate network using software-defined networking (SDN), virtualization, tunneling, and/or encapsulation technologies. An overlay network generally operates independently of the underlying substrate network. An overlay network may provide logical separation and isolation of traffic, enable virtual network provisioning, and/or allow for implementation of various network services and policies. Virtual machines, hosts, containers, or virtual network functions running on a substrate network may be connected via an overlay network.

(150) As used herein, the term “overlay entity” refers to a network entity **658** implemented on an overlay network **654**. The overlay network **654** may include a plurality of overlay entities **664**. The plurality of overlay entities **664** may include overlay hosts, overlay services, subnets, overlay controllers, and/or overlay clients. In one example, the overlay network **654** may include a plurality of overlay entities **664**, such as overlay entity **664c** and overlay entity **664n**. In one example, an overlay entity **664** may include an overlay host. Additionally, or in the alternative, an overlay entity **664** may include an overlay service. The plurality of overlay entities **664** may communicate with one another using logical network addresses assigned within the overlay network **654**.

(151) An overlay host may include an endpoint within the overlay network **654**, such as a virtual machine, a container, or a physical server. An overlay service may include a service executing or executable on an overlay entity. An overlay service may include a client-specific service, such as a service installed by a client. Additionally, or in the alternative, an overlay service may include a virtual network creation service, a virtual network management service, a virtual machine orchestration service, a container orchestration service, a network virtualization service, an overlay security service, a load balancing service, a multi-tenancy service, and/or a tenant isolation service.

(152) A subnet may include a virtual network segment that has a distinct addressing scheme and/or a distinct set of network policies and/or services. A subnet may include a set of overlay hosts. Multiple subnets may be utilized to partition respective sets of overlay hosts. An overlay controller may oversee management, control, provisioning, configuration, and/or monitoring of an overlay network, network entities on the overlay network, and/or network policies within the overlay. An overlay controller interact with the underlying substrate network, for example, to coordinate the operation of overlay hosts and/or communications across virtual switches and tunnels. An overlay client may include an endpoint or device that initiates communication within the overlay network. An overlay client may be a specific instance or role within an overlay host. An overlay host may include a set of overlay clients. An overlay client may include a consumer or user of services provided by overlay hosts or the IaaS. An overlay client may request and consume resources or services from overlay hosts, acting as consumers or clients of those resources or services.

(153) Referring further to FIG. 6D, the system **600** may include at least one distribution service **666** and at least one certificate repository **668**. The at least one distribution service **666** may correspond to the distribution service **602** described with reference to FIG. 6A. A distribution service **666** may be configured to distribute certificate bundles that include CA certificates to at least some of the network entities **658**. The certificate bundles may be housed in a certificate repository **668** accessible by the distribution service **666**. The certificate repository **668** may correspond to the certificate repository **604** described with reference to FIG. 6A. The distribution service **666** may retrieve a certificate bundle from the certificate repository **668** and transmit the certificate bundle to one or more of the network entities **658**. In one example, a distribution service **666** may be configured to transition, such as concurrently transition, a plurality of network entities through a series of stages of a certificate bundle distribution process for distributing a new set of one or more CA certificates to the plurality of network entities **658** for use in a certificate authentication process.

(154) In one example, a first distribution service **666a** may be implemented on the substrate network **652**. A distribution service **666** implemented on the substrate network **652** may be communicatively coupled with one or more substrate entities **660** on the substrate network **652**.

Additionally, or in the alternative, a distribution service **666** implemented on the substrate network **652** may be communicatively coupled with the network interface **656**, such as with one or more interface entities **662** on the network interface **656**. For example, the first distribution service **666a** may obtain certificate bundles from the certificate repository **668** and distribute the certificate bundles to one or more substrate entities **660** on the substrate network **652** and/or to one or more interface entities **662** on the network interface **656**.

(155) In one example, a second distribution service **666b** may be implemented on the network interface **656**. A distribution service **666** implemented on the network interface **656** may be communicatively coupled with the overlay network **654** and/or the substrate network **652**. For example, the second distribution service **666b** implemented on the network interface **656** may be communicatively coupled with one or more overlay entities **664** on the overlay network **654**. The second distribution service **666b** may distribute certificate bundles to one or more overlay entities **664** on the overlay network **654**. In one example, the second distribution service **666b** may obtain certificate bundles from a storage medium associated with the network interface **656**, such as from a storage medium associated with an interface entity **612** on the network interface **656**. For example, the second distribution service **666b** may distribute a certificate bundle that has been installed in the storage medium associated with the network interface **656** and/or the interface entity **662**. In one example, the first distribution service **666a** may distribute a certificate bundle to an interface entity **662** on the network interface **656**, and the second distribution service **666b** may distribute the certificate bundle to one or more overlay entities **664** on the overlay network **654** after the certificate bundle has been installed in the storage medium associated with the interface entity **662**.

(156) As shown in FIG. 6D, each network entity **658** may include a provisioning module **670** and a storage medium **672**. The provisioning module **670** may correspond to provisioning module **618** described with reference to FIG. 6A. The storage medium **672** may correspond to the storage medium **620** described with reference to FIG. 6A. The provisioning module **670** associated with a network entity **658** may receive certificate bundles from the distribution service **666** and install the certificate bundle in the storage medium **672** associated with the network entity **658**. The distribution service **666** may distribute a certificate bundle to the provisioning module **670** in connection with an initial provisioning of the network entity **658** and/or in connection with a periodic update. Additionally, or in the alternative, the distribution service **666** may distribute a certificate bundle to the provisioning module **670** in accordance with a respective stage of a certificate bundle distribution process.

(157) In one example, a provisioning module **670** of an overlay entity **664** may send requests for a certificate bundle to a distribution service **666** by way of a gateway **674** between the network interface **656** and the overlay entities **664**. Additionally, or in the alternative, the distribution service **666** may send the certificate bundles to the overlay entities **664** by way of the gateway **674**, for example, in response to a request from a provisioning module **670**. In one example, the gateway **674** may be a metadata interface configured for exchanging metadata between the network interface **656** and the overlay entities **664**. In one example, the gateway **674** may be utilized between the network interface **656** and each of the overlay entities **664**. Additionally, or in the alternative, the network interface **656** may include a set of gateways **674**, with each gateway **674** corresponding to a respective overlay entity **664**.

(158) In one or more embodiments, the system **600** may include more or fewer components than the components illustrated in FIGS. 6A-6D. The components illustrated in FIGS. 6A-6D may be local to or remote from each other. The components illustrated in FIGS. 6A-6D may include software and/or hardware components. Each component may be distributed over multiple applications and/or machines. Multiple components may be combined into one application and/or machine. Operations described with respect to one component may instead be performed by another component. Additional embodiments and/or examples relating to the system **600** are

described above in Section 2, titled “Cloud Computing Technology”.

(159) In an embodiment, the system **600** may include various components implemented on one or more digital devices. The term “digital device” generally refers to any hardware device that includes a processor. A digital device may refer to a physical device executing an application or a virtual machine. Examples of digital devices include a computer, a tablet, a laptop, a desktop, a netbook, a server, a web server, a network policy server, a proxy server, a generic machine, a function-specific hardware device, a hardware router, a hardware switch, a hardware firewall, a hardware firewall, a hardware network address translator (NAT), a hardware load balancer, a mainframe, a television, a content receiver, a set-top box, a printer, a mobile handset, a smartphone, a personal digital assistant (PDA), a wireless receiver and/or transmitter, a base station, a communication management device, a router, a switch, a controller, an access point, and/or a client device.

5. Example Certificate Bundle Distribution Process

(160) Referring now to FIG. 7, operations pertaining to an example certificate bundle distribution process are further described. The operations **700** described with reference to FIG. 7 may be associated with distributing a new set of one or more CA certificates to a plurality of network entities for use in a certificate authentication process. The CA certificates may include root CA certificates and/or intermediate CA certificates. The CA certificates may be housed in a certificate bundle. One or more operations **700** illustrated in FIG. 7 may be modified, rearranged, or omitted all together. Accordingly, the particular sequence of operations **700** illustrated in FIG. 7 should not be construed as limiting the scope of one or more embodiments. As shown in FIG. 7, operations **700** pertaining to the certificate bundle distribution process may include a series or sequence of stages. In one example, as shown, the operations **700** may include a series or sequence of stages, such as four (4) stages. In another example, the series or sequence of stages may include more or less than four (4) stages, for example, including all or a portion of the stages shown in FIG. 7.

(161) In one example, the operations **700** may include, at block **702**, distributing a new certificate bundle including a new set of one or more CA certificates to a set of network entities associated with a virtual cloud network. The new set of one or more CA certificates may include at least one new CA certificate. Additionally, or in the alternative, a set of new CA certificates may be added to existing certificate bundles.

(162) At block **704**, the operations **700** may include determining whether the new certificate bundle has been successfully distributed. When the new certificate bundle has been successfully distributed, the operations **700** may proceed to a next stage in the distribution process. The operation at block **702** and/or **704** may represent Stage 1 of a certificate bundle distribution process.

(163) In one example, the operations **700** may include, at block **706**, activating the CA certificates in the new certificate bundle. In one example, activating the CA certificate may include a CA corresponding to the CA certificate issuing one or more intermediate CA certificates. The operation at block **706** may represent Stage 2 of a certificate bundle distribution process.

(164) In one example, the operations **700** may include, at block **708**, distributing new entity certificates to the set of network entities. Additionally, or in the alternative, the operations **700** may include distributing new intermediate CA certificates, for example, corresponding to the new entity certificates.

(165) At block **710**, the operations **700** may include determining whether the new entity certificates have been successfully distributed. Additionally, or in the alternative, the operations may include determining whether the new intermediate CA certificates have been successfully distributed. When the new entity certificates (and/or the new intermediate CA certificates) have been successfully distributed, the operations **700** may proceed to a next stage in the distribution process. The operation at block **708** and **710** may represent Stage 3 of a certificate bundle distribution process.

(166) In one example, the operations **700** may include, at block **712**, removing earlier CA certificates that were distributed prior to the new CA certificates. For example, the earlier CA certificates may include CA certificates that are being superseded by the new CA certificates. Removing an earlier CA certificate with respect to a particular network entity may include deleting the earlier CA certificate from a storage medium associated with the particular network entity. For example, the operations **700** may include transmitting, to the particular network entity, an instruction to delete the earlier CA certificate from the storage medium. Responsive to receiving the instruction, the particular network entity may delete the earlier CA certificate from the storage medium.

(167) Additionally, or in the alternative, removing an earlier CA certificate with respect to a network entity may include replacing a first new certificate bundle with a second new certificate bundle, in which the first new certificate bundle includes the new CA certificates and the earlier CA certificates and the second new certificate bundle includes only the new CA certificates, thereby effectively removing the earlier CA certificates from the first new certificate bundle. The second new certificate bundle may be distributed to the particular network entity and installed in the storage medium associated with the particular network entity, and the first new certificate bundle may be deleted from the storage medium. The operation at block **712** may represent Stage 4 of a certificate bundle distribution process.

6. Example Certificate Bundle Distribution Operations

(168) Referring now to FIGS. **8A** and **8B**, operations **800** pertaining to distributing certificate bundles are further described. The operations **800** described with reference to FIGS. **8A** and **8B** may represent at least a portion of a process for distributing certificate bundles to network entities, for example, as described herein with reference to FIG. **7**. One or more operations **800** described with reference to in FIGS. **8A** and **8B** may be modified, combined, rearranged, or omitted all together. Accordingly, the particular sequence of operations **800** described with reference to FIGS. **8A** and **8B** should not be construed as limiting the scope of one or more embodiments.

A. Distributing Certificate Bundles According to Certificate Distribution Groups or Subgroups

(169) Referring to FIG. **8A**, operations associated with distributing certificate bundles according to certificate distribution groups are further described. As shown in FIG. **8A**, the operations **800** may include, at block **802**, detecting a trigger condition to distribute a certificate bundle that includes one or more CA certificates.

(170) At block **804**, the operations **800** may include determining, for each particular network entity of a set of network entities, a fault domain associated with the particular network entity.

(171) In one example, the operations **800** may include, at block **804**, determining, for each particular network entity of a subset of network entities associated with a certificate distribution group, a particular certificate distribution subgroup, from among a plurality of certificate distribution subgroups, based on a network address of the particular network entity. Each particular certificate distribution subgroup may include a subgroup of network entities from the subset of network entities in the certificate distribution group from which the subgroup of network entities are assigned. The particular certificate distribution subgroup corresponding to a particular network entity may be determined based on the network address of the particular network entity.

(172) At block **806**, the operations **800** may include partitioning the set of network entities into a plurality of certificate distribution groups based on a set of one or more partitioning criteria. The set of one or more partitioning criteria may include the fault domain associated with the particular network entity. Additionally, or in the alternative, the set of one or more partitioning criteria may include a resource group corresponding to the particular network entity.

(173) At block **808**, the operations **800** may include selecting one of the certificate distribution groups, from the plurality of certificate distribution groups, for distribution of the certificate bundle based on the particular release phase corresponding to the certificate distribution group having commenced. In one example, the operations **800** may include, at block **808**, selecting one of the

certificate distribution subgroups, from the selected certificate distribution group, for distribution of the certificate bundle based on the particular release phase corresponding to the certificate distribution subgroup having commenced.

(174) At block **810**, the operations **800** may include transmitting the certificate bundle to each of the network entities in the selected certificate distribution group. In one example, the operations **800** may include, at block **810**, transmitting the certificate bundle to each of the network entities in the selected certificate distribution subgroup.

(175) At block **812**, the operations **800** may include determining whether there is another certificate distribution group, from the plurality of certificate distribution group, that includes a subset of network entities that have yet to receive the certificate bundle. If there is another certificate distribution group, the operations **800** may proceed to block **808**, where another certificate distribution group may be selected. When the certificate bundle has been transmitted to each subset of network entities, such that, at block **812**, there is not another certificate distribution groups, the operations may end at block **814**.

(176) In one example, the operations **800** may include, at block **812**, determining whether there is another certificate distribution subgroup, from a particular certificate distribution group, that includes a subgroup of network entities that have yet to receive the certificate bundle. If there is another certificate distribution subgroup in the certificate distribution group, the operations **800** may include proceeding to distribute the certificate bundle to the next certificate distribution subgroup. When the certificate bundle has been transmitted to each subgroup of network entities, such that, at block **812**, there is not another certificate distribution subgroup in the certificate distribution group, the operations **800** may include proceeding to distribute the certificate bundle to the next certificate distribution group.

B. Requests from Network Entities for Certificate Bundle Distribution

(177) Referring to FIG. **8B**, operations associated with distributing certificate bundles in response to requests from network entities are further described. As shown in FIG. **8B**, the operations **800** may include, at block **822** receiving, from a network entity associated with a virtual cloud network, a request for certificate bundle distribution.

(178) At block **824**, the operations **800** may include determining, based at least in part on a distribution schedule for distributing a new certificate bundle within the virtual cloud network, a certificate distribution group (and/or a certificate distribution subgroup) corresponding to the network entity and a release phase for releasing the new certificate bundle for distribution to the certificate distribution group (and/or to a certificate distribution subgroups corresponding to the certificate distribution group). The distribution schedule may include a set of certificate distribution groups (and/or a set of certificate distribution subgroup corresponding to a certificate distribution group) and a set of release phases corresponding to the set of certificate distribution groups (and or to the set of certificate distribution subgroups). Each particular release phase of the set of release phases may correspond to at least one certificate distribution group of the set of certificate distribution groups. Each particular certificate distribution group may correspond to one or more partitioning criteria, such as a fault domain and/or a resource group. In one example, each particular release phase of the set of release phases may correspond to at least one certificate distribution subgroup of a particular certificate distribution group. Each particular certificate distribution group may correspond to a particular set of network addresses of the virtual cloud network.

(179) At block **826**, the operations **800** may include determining whether the release phase corresponding to the certificate distribution group (and/or to the certificate distribution subgroup) has commenced prior to the request from the network entity. If the release phase has commenced prior to the request from the network entity, the operations **800** may proceed to block **828**. If the release phase has not commenced prior to the request from the network entity, the operations **800** may proceed to block **832**.

(180) At block **828**, when the release phase has commenced prior to the request from the network entity, the operations **800** may include selecting the new certificate bundle for distribution to the network entity. The new certificate bundle may be selected for distribution to the network entity based at least in part on the release phase having commenced prior to the request from the network entity. At block **830**, the operations **800** may include transmitting the new certificate bundle to the network entity. The new certificate bundle may include one or more CA certificates. The one or more CA certificates may include at least one new CA certificate. Following the operations **800** at block **830**, operations **800** may return to block **822**, where another request for certificate bundle distribution may be received from another network entity.

(181) At block **832**, when the release phase has not commenced prior to the request from the network entity, the operations **800** may include selecting an earlier certificate bundle for distribution to the network entity. The earlier certificate bundle may be selected for distribution to the network entity based at least in part on the release phase having yet to commence prior to the request from the network entity. At block **830**, the operations **800** may include transmitting the earlier certificate bundle to the network entity. The earlier certificate bundle may include one or more CA certificates. The one or more CA certificates may include at least one earlier CA certificate. Following the operations **800** at block **834**, operations **800** may return to block **822**, where another request for certificate bundle distribution may be received from another network entity.

(182) In one example, the operations **800** described with reference to FIGS. **8A** and **8B** may be combined. For example, determining the certificate distribution groups based on the one or more partitioning criteria, at block **804** of FIG. **8A**, may include receiving a request for certificate bundle distribution from a network entity at block **822** of FIG. **8B**. Additionally, or in the alternative, determining the certificate distribution subgroups based on the network address of the particular network entity, at block **804** of FIG. **8A**, may include receiving a request for certificate bundle distribution from a network entity at block **822** of FIG. **8B**. As another example, determining the certificate distribution groups based on the one or more partitioning criteria, at block **804** of FIG. **8A**, may include determining, based at least in part on the distribution schedule for distributing the certificate bundle, a certificate distribution group corresponding to the network entity and a release phase for releasing the certificate bundle for distribution to the certificate distribution group, at block **824** of FIG. **8B**. Additionally, or in the alternative, determining the certificate distribution subgroups based on the network address of the particular network entity, at block **804** of FIG. **8A**, may include determining, based at least in part on the distribution schedule for distributing the certificate bundle, a certificate distribution group corresponding to the network entity and a release phase for releasing the certificate bundle for distribution to the certificate distribution group, at block **824** of FIG. **8B**.

C. Distribution Criteria for Transmitting Certificate Bundles

(183) In one example, the operations **800** may include progressing through a series of certificate distribution groups (and/or certificate distribution subgroups) in a distribution schedule based at least in part on one or more distribution criteria being satisfied for a current certificate distribution group (and/or for a certificate distribution subgroups). For example, the operations **800** may include transmitting the certificate bundle to each of the network entities in the selected certificate distribution group (and/or the selected certificate distribution subgroup), for example, at block **810** of FIG. **8A**, determining that one or more distribution criteria are satisfied with respect to the selected certificate distribution group (and/or the selected certificate distribution subgroup), and responsive to determining that the one or more distribution criteria are satisfied, transmitting the certificate bundle to the network entities in a next certificate distribution group (and/or the next certificate distribution subgroup).

(184) Additionally, or in the alternative, the operations **800** may include progressing to through a series of release phases in a distribution schedule based at least in part on one or more distribution

criteria being satisfied for a release phase. For example, with reference to FIG. 8B, the operations **800** may include commencing a release phase for releasing a certificate bundle for distribution to a certificate distribution group (and/or a certificate distribution subgroup), determining that one or more distribution criteria are satisfied, and commencing a next release phase in the distribution schedule responsive to determining that one or more distribution criteria are satisfied.

(185) In one example, with reference to FIG. 8A and/or FIG. 8B, the operations **800** may include determining a distribution metric with respect to distribution of a certificate bundle to a first subset of network entities corresponding to a first certificate distribution group (and/or a first certificate distribution subgroup), and determining whether the distribution metric meets one or more distribution criteria. Responsive to the distribution metric meeting the one or more distribution criteria, the operations **800** may include transmitting the certificate bundle to a second subset of network entities corresponding to a second certificate distribution group (and/or a second certificate distribution subgroup). In one example, with reference to FIG. 8A, the operations **800** may include, prior to proceeding from block **812** to block **808**, determining whether the distribution metric meets one or more distribution criteria. If the distribution metric meets the one or more distribution criteria, the operations **800** may proceed to block **808**.

(186) In one example, the distribution metric may include an error count associated with transmitting a certificate bundle to a subset of network entities corresponding to a certificate distribution group (and/or a certificate distribution subgroup). The error count may be indicative of a number or a proportion of network entities from among the subset of network entities with respect to which an error event associated with the certificate bundle occurs during a verification period. The one or more distribution criteria may include the error count remaining below a threshold during the verification period.

(187) In one example, the distribution metric may include a distribution count associated with transmitting a certificate bundle to a subset of network entities corresponding to a certificate distribution group (and/or a certificate distribution subgroup). The distribution count may be indicative of a number or a proportion of network entities from among the subset of network entities with respect to which a distribution indicator indicates a successful distribution of the certificate bundle. The one or more distribution criteria may include the distribution count meeting a threshold.

D. Further Example Partitioning Criteria

(188) Referring further to FIGS. 8A and 8B, in one example, the plurality of network entities may be assigned to one of a plurality of certificate distribution groups based on one or more partitioning criteria. The one or more partitioning criteria may include a fault domain, a resource group, and/or an availability domain, as described above. Additionally, or in the alternative, a subset of network entities of a certificate distribution group may be assigned to one of a plurality of certificate distribution subgroups based on a network address of the particular network entity, as described above.

(189) In one example, one or more of the certificate distribution groups (and/or one or more of the certificate distribution subgroups) may be further assigned to one or more certificate distribution subsets based at least in part on one or more successful distribution variables and/or one or more unsuccessful distribution variables. In one example, each particular network entity corresponding to a particular certificate distribution subset may be associated with a particular successful distribution variable and/or a particular unsuccessful distribution variable corresponding to the particular certificate distribution subgroup. The one or more successful distribution variables may include one or more characteristics associated with a likelihood of having a successful distribution. The one or more unsuccessful distribution variables may include one or more characteristics associated with a likelihood of an unsuccessful distribution. In one example, the operations **800** may include progressing through a series of certificate distribution subsets of a certificate distribution group (and/or a certificate distribution subgroup) based at least in part on one or more

distribution criteria pertaining to a successful distribution variable and/or an unsuccessful distribution variable being satisfied for a current certificate distribution subset. For example, the operations **800** may include transmitting the certificate bundle to each of the network entities in a selected certificate distribution subset, determining that one or more distribution criteria pertaining to a successful distribution variable and/or an unsuccessful distribution variable are satisfied with respect to the selected certificate distribution subset, and responsive to determining that the one or more distribution criteria are satisfied, transmitting the certificate bundle to the network entities in a next certificate distribution subset.

E. Distribution Rules for Interface Entities

(190) Referring further to FIGS. **8A** and **8B**, in one example, a network entity that is an interface entity may be assigned to a certificate distribution group (and/or a certificate distribution subgroup) based on one or more distribution rules. In one example, a distribution rule may augment the certificate distribution group to which a network entity is assigned. For example, the distribution rule may move interface entities that support a large number of overlay entities to a later portion of a distribution schedule. The distribution rule may serve to prevent an unsuccessful distribution to an interface entity from occurring early in the distribution schedule, particularly in the event that the interface entity supports a relatively large number of overlay entities. In one example, one or more network entities may be assigned to an aft-distribution group based on one or more distribution rules.

(191) In one example, a first certificate distribution group (and/or a first certificate distribution subgroup) may be determined for a network entity, for example, based on one or more partitioning criteria, and as second certificate distribution group (and/or a second certificate distribution subgroup) may be determined for the network entity based on the one or more distribution rules. A first release phase corresponding to the first distribution group and second release phase corresponding to the second distribution group may be determined, and the network entity may be assigned to the distribution group corresponding to the later occurring of the first and second release phases.

7. Authenticating Network Entities

(192) Network entities may utilize the CA certificates in a certificate bundle to authenticate other network entities associated with the virtual cloud network. For example, communications between network entities may be conducted according to a security protocol. The security protocol may include authenticating a network entity based on an entity certificate issued to the network entity by a CA, for example, prior to establishing communications with the network entity.

(193) In one example, the entity certificate and a CA certificate corresponding to the CA that issued the entity certificate may represent at least a portion of a certificate chain. To authenticate the network entity, each signature-key pair in the certificate chain is validated. In one example, a top-level CA may issue the entity certificate, in which case the certificate chain may include one signature-key pair—that is, the digital signature of the top-level CA in the entity certificate, and the public key of the top-level CA. Such a top-level CA is sometimes referred to as a root CA. In another example, the certificate chain may include signature-key pairs corresponding to multiple CA certificates. For example, a root CA may issue an intermediate CA certificate to an intermediate CA, and the intermediate CA may issue the entity certificate to the network entity. In this case, the certificate chain includes two signature-key pairs—that is, (i) the digital signature of the intermediate CA in the entity certificate, and the public key of the intermediate CA; and (ii) the digital signature of the root CA in the intermediate CA certificate, and the public key of the root CA.

(194) As used herein, the term “certificate authority certificate” or “CA certificate” refers to a digital certificate issued by a CA to establish its own identity and authenticity. A certificate authority certificate may be a root CA certificate or an intermediate CA certificate. A certificate authority certificate may be used to sign and issue other digital certificates, including those used for

secure communication between network entities.

(195) As used herein, the term “certificate authority” or “CA” refers to an entity responsible for issuing and managing digital certificates. The CA verifies the identity of network entities and digitally signs their certificates to attest to their authenticity.

(196) As used herein, the term “root certificate authority certificate” or “root CA certificate” refers to a top-level CA certificate in a certificate chain or hierarchy. A root CA certificate may be self-issued and/or self-signed by a root CA. As used herein, the term “root CA” refers to a top-level CA in a CA hierarchy. A root CA may issue root CA certificates, intermediate CA certificates, or entity certificates.

(197) As used herein, the term “intermediate certificate authority certificate” or “intermediate CA certificate” refers to an intermediate-level CA certificate in a certificate chain or hierarchy. An intermediate CA certificate may be issued by a root CA. An intermediate CA certificate is located between a root CA certificate and an entity certificate in a certificate chain or hierarchy. As used herein, the term “intermediate CA” refers to an intermediate-level CA in a CA hierarchy. An intermediate CA may issue entity certificates, for example, pursuant to authority granted to an intermediate CA according to a root CA.

(198) As used herein, the term “entity certificate” refers to a digital certificate issued to an entity, such as a network entity associated with a virtual cloud network. An entity certificate may be used to verify the identity of the entity and enable secure communication between entities, such as between network entities in a virtual cloud network. An entity certificate may be issued by a CA, such as root CA or an intermediate CA.

(199) In one example, an entity certificate may be an instance principal certificate. As used herein, the term “instance principal certificate” refers to a digital certificate used to authenticate and secure communication for an instance or VM associated with a virtual cloud network. In one example, instances and VMs may be created, scaled, and terminated dynamically. Instance principal certificates may be associated with an instance or VAM during its lifecycle and may be automatically generated and managed by the virtual cloud network infrastructure. An instance principal certificate may have limited access to communicate with certain network entities based on permissions assigned to the network entity to which the instance principal certificate is issued.

(200) As used herein, the term “digital certificate” refers to a digitally signed electronic document that binds a public key to the identity of an entity. A digital certificate may conform to International Telecommunication Union standard X.509. A digital certificate may include an issuer's name, a certificate holder's name, a public key, issuer (CA) information, and expiration date. Digital certificates may be used in various security protocols, such as SSL/TLS, to establish the identity and authenticity of the communicating parties and facilitate secure communication.

8. Miscellaneous; Extensions

(201) Embodiments are directed to a system with one or more devices that include a hardware processor and that are configured to perform any of the operations described herein and/or recited in any of the claims below. Embodiments are directed to a system including means to perform any of the operations described herein and/or recited in any of the claims below. In an embodiment, a non-transitory computer readable storage medium comprises instructions which, when executed by one or more hardware processors, causes performance of any of the operations described herein and/or recited in any of the claims.

(202) Further aspects of the presently disclosed subject matter are provided by the following clauses:

(203) One or more non-transitory computer readable media comprising instructions which, when executed by one or more hardware processors, causes performance of operations comprising: receiving, from a first network entity associated with a computer network, a first request for certificate bundle distribution; determining, based at least in part on a distribution schedule for distributing a first certificate bundle within the computer network, a first certificate distribution

group corresponding to the first network entity and a first release phase for releasing the first certificate bundle for distribution to the first certificate distribution group, wherein the distribution schedule comprises (a) a set of certificate distribution groups, including the first certificate distribution group, and (b) a set of release phases, including the first release phase, for releasing the first certificate bundle for distribution to particular certificate distribution groups of the set of certificate distribution groups, wherein each particular certificate distribution group of the set of certificate distribution groups corresponds to a particular subset of network entities of a plurality of network entities associated with the computer network, wherein the particular subset of network entities meet a set of one or more partitioning criteria, wherein the set of one or more partitioning criteria comprises a fault domain of the particular network entity, wherein the computer network comprises a plurality of fault domains respectively representing a single point of failure, wherein the particular subset of network entities are associated with a particular single point of failure corresponding to a particular fault domain of the plurality of fault domains, and wherein the first certificate distribution group, including the first network entity, corresponds to a first fault domain of the plurality of fault domains, wherein each particular release phase of the set of release phases corresponds to at least one certificate distribution group of the set of certificate distribution groups, and wherein the first release phase has commenced prior to having received the first request; selecting the first certificate bundle for distribution to the first network entity based at least in part on the first release phase having commenced prior to receiving the first request, wherein the first certificate bundle comprises a first set of one or more certificate authority (CA) certificates; transmitting the first certificate bundle to the first network entity.

(204) The media of any clause herein, wherein determining the first certificate distribution group corresponding to the first network entity comprises: determining the first fault domain based on a first metadata element associated with the first network entity, wherein the first metadata element comprises a first fault domain attribute, wherein the first fault domain attribute identifies the first network entity as corresponding to the first fault domain.

(205) The media of any clause herein, wherein the set of one or more partitioning criteria comprises a resource group of the particular network entity, wherein the plurality of network entities are associated with one of a plurality of resource groups respectively representing at least one of: an attribute, a functionality, or a purpose, wherein the particular subset of network entities are associated a particular resource group, of the plurality of resource groups, representing at least one of: a particular attribute, a particular functionality, or a particular purpose, and wherein the first certificate distribution group, including the first network entity, corresponds to a first resource group of the plurality of resource groups.

(206) The media of any clause herein, wherein determining the first certificate distribution group corresponding to the first network entity comprises: determining the first resource group based on a first metadata element associated with the first network entity, wherein the first metadata element comprises a first resource group attribute, wherein the first resource group attribute identifies the first network entity as corresponding to the first resource group.

(207) The media of any clause herein, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first network entity, receiving, from a second network entity associated with the computer network, a second request for certificate bundle distribution; determining, based at least in part on the distribution schedule for distributing the first certificate bundle within the computer network, a second certificate distribution group corresponding to the second network entity and a second release phase for releasing the first certificate bundle for distribution to the second certificate distribution group, wherein the second certificate distribution group, including the second network entity, corresponds to a second fault domain of the plurality of fault domains, and wherein the second release phase has commenced prior to having received the second request; selecting the first certificate bundle for distribution to the second network entity based at least in part on the second release phase having commenced prior to receiving the second

request; and transmitting the first certificate bundle to the second network entity.

(208) The media of any clause herein, wherein the second certificate distribution group corresponds to the first resource group of the plurality of resource groups.

(209) The media of any clause herein, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first network entity, determining a distribution metric with respect to distribution of the first certificate bundle to the first certificate distribution group during the first release phase; determining that the distribution metric meets a distribution criterion; and responsive to the distribution metric meeting the distribution criterion, commencing a second release phase for releasing the first certificate bundle for distribution to a second certificate distribution group.

(210) The media of any clause herein, wherein the distribution metric comprises an error count associated with transmitting the first certificate bundle to a first subset of network entities of the first certificate distribution group, the error count indicative of a number or a proportion of network entities from among the first subset of network entities with respect to which an error event associated with the first certificate bundle occurs during a verification period; and wherein the distribution criterion comprises the error count remaining below a threshold during the verification period.

(211) The media of any clause herein, wherein the distribution metric comprises a distribution count associated with transmitting the first certificate bundle to a first subset of network entities of the first certificate distribution group, the distribution count indicative of a number or a proportion of network entities from among the first subset of network entities with respect to which a distribution indicator indicates a successful distribution of the first certificate bundle; and wherein the distribution criterion comprises the distribution count meeting a threshold.

(212) The media of any clause herein, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first network entity, receiving, from a second network entity associated with the computer network, a second request for certificate bundle distribution; determining, based at least in part on the distribution schedule for distributing the first certificate bundle within the computer network, a second certificate distribution group corresponding to the second network entity and a second release phase for releasing the first certificate bundle for distribution to the second certificate distribution group, wherein the second certificate distribution group, including the second network entity, corresponds to a second resource group of the plurality of resource groups, and wherein the second release phase has commenced prior to having received the second request; selecting the first certificate bundle for distribution to the second network entity based at least in part on the second release phase having commenced prior to receiving the second request; and transmitting the first certificate bundle to the second network entity.

(213) The media of any clause herein, wherein the second certificate distribution group corresponds to the first fault domain of the plurality of fault domains.

(214) The media of any clause herein, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the second network entity, receiving, from a third network entity associated with the computer network, a third request for certificate bundle distribution; determining, based at least in part on the distribution schedule for distributing the first certificate bundle within the computer network, a third certificate distribution group corresponding to the third network entity and a third release phase for releasing the first certificate bundle for distribution to the third certificate distribution group, wherein the third certificate distribution group, including the third network entity, corresponds to a second fault domain of the plurality of fault domains, and wherein the third release phase has commenced prior to having received the third request; selecting the first certificate bundle for distribution to the third network entity based at least in part on the third release phase having commenced prior to receiving the third request; and transmitting the first certificate bundle to the third network entity.

(215) The media of any clause herein wherein the third certificate distribution group corresponds to

the first resource group of the plurality of resource groups.

(216) The media of any clause herein, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first network entity: detecting a trigger condition to commence a second release phase for releasing the first certificate bundle for distribution to a second certificate distribution group, wherein the trigger condition comprises at least one of: a time interval having elapsed subsequent to transmitting the first certificate bundle to a first subset of network entities, including the first network entity, of the first certificate distribution group, or an update to a release phase attribute in a configuration dataset, the update to the release phase attribute indicating to commence the second release phase; and responsive to detecting the trigger condition, commencing the second release phase.

(217) The media of any clause herein, wherein the operations further comprise: receiving, from a second network entity associated with the computer network, a second request for certificate bundle distribution; determining, based at least in part on the distribution schedule, a second certificate distribution group corresponding to the second network entity and a second release phase for releasing the first certificate bundle for distribution to the second certificate distribution group, wherein the set of certificate distribution groups includes the second certificate distribution group, and wherein the set of release phases includes the second release phase, and wherein the second release phase has yet to commence; selecting a second certificate bundle for distribution to the first network entity, wherein the second certificate bundle is selected in lieu of the first certificate bundle based at least in part on the second release phase having yet to commence, wherein the second certificate bundle comprises a second set of one or more CA certificates, wherein the second set of one or more CA certificates differs from the first set of one or more CA certificates; and transmitting the second certificate bundle to the second network entity.

(218) The media of any clause herein, wherein the set of one or more partitioning criteria comprises a randomization parameter determined by applying a randomization function to a particular network address of the particular network entity, wherein determining the first certificate distribution group comprises applying the randomization function to a first network address of the first network entity.

(219) The media of any clause herein, wherein the operations further comprise: prior to receiving the first request for certificate bundle distribution, detecting a trigger condition to distribute the first certificate bundle; and responsive to detecting the trigger condition, commencing the first release phase for releasing the first certificate bundle for distribution to the first certificate distribution group.

(220) The media of any clause herein, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first network entity, detecting a trigger condition to distribute a second certificate bundle comprising a second set of one or more CA certificates, wherein the second set of one or more CA certificates differs from the first set of one or more CA certificates; responsive to detecting the trigger condition, commencing a second release phase, of the set of release phases in the distribution schedule, for releasing the second certificate bundle for distribution to a second certificate distribution group; receiving, from a second network entity associated with the computer network, a second request for certificate bundle distribution; determining, based at least in part on a second distribution schedule for distributing the second certificate bundle within the computer network, that the second network entity corresponds to the second certificate distribution group, wherein the second certificate distribution group corresponds to the second release phase, and wherein the second release phase has commenced prior to having received the second request; selecting the second certificate bundle for distribution to the second network entity based at least in part on the second release phase having commenced prior to receiving the second request; and transmitting the second certificate bundle to the second network entity.

(221) The media of any clause herein, wherein the operations further comprise: receiving, from a

third network entity associated with the computer network, a third request for certificate bundle distribution; determining, based at least in part on the distribution schedule, a third certificate distribution group corresponding to the third network entity and a third release phase for releasing the second certificate bundle for distribution to the third certificate distribution group, wherein the set of certificate distribution groups includes the third certificate distribution group, and wherein the set of release phases includes the third release phase, and wherein the third release phase has yet to commence; selecting the first certificate bundle for distribution to the third network entity, wherein the first certificate bundle is selected in lieu of the second certificate bundle based at least in part on the third release phase having yet to commence; and transmitting the first certificate bundle to the second network entity.

(222) Any combination of the features and functionalities described herein may be used in accordance with one or more embodiments. In the foregoing specification, embodiments have been described with reference to numerous specific details that may vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the invention, and what is intended by the applicants to be the scope of the invention, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction.

(223) All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

Claims

1. One or more non-transitory computer readable media comprising instructions which, when executed by one or more hardware processors, causes performance of operations comprising: detecting a trigger condition to distribute a first certificate bundle comprising a first set of one or more certificate authority certificates; determining, for each particular network entity of a plurality of network entities associated with a computer network, a fault domain associated with the particular network entity, from among a plurality of fault domains associated with the computer network, wherein each particular fault domain, of the plurality of fault domains, represents at least one single point of failure; partitioning the plurality of network entities into a plurality of certificate distribution groups based on a set of one or more partitioning criteria, wherein the set of one or more partitioning criteria comprises the fault domain associated with the particular network entity, and wherein each particular certificate distribution group, of the plurality of certificate distribution groups, comprises a particular subset of network entities, of the plurality of network entities, wherein the particular subset of network entities is associated with a particular single point of failure corresponding to a particular fault domain of the plurality of fault domains; selecting a first certificate distribution group, of the plurality of certificate distribution groups, for distribution of the first certificate bundle, wherein the first certificate distribution group comprises a first subset of network entities that is associated with a first fault domain of the plurality of fault domains; transmitting the first certificate bundle to the first subset of network entities.
2. The media of claim 1, wherein the first subset of network entities is not associated with a second fault domain of the plurality of fault domains.
3. The media of claim 2, wherein at least a first network entity, of the first subset of network entities, performs a first service with respect to the first fault domain, and wherein at least a second network entity, of a second subset of network entities, performs the first service with respect to the second fault domain.
4. The media of claim 1, wherein the fault domain of each particular network entity is determined based on metadata associated with each particular network entity, wherein the metadata comprises

a fault domain attribute identifying the fault domain corresponding to the particular network entity, wherein for a first network entity of the first subset of network entities, a first metadata element comprises a first fault domain attribute, wherein the first fault domain attribute identifies the first network entity as corresponding to the first fault domain.

5. The media of claim 1, wherein the operations further comprise: determining, for each particular network entity of the plurality of network entities, a resource group associated with the particular network entity, from among a plurality of resource groups associated with the computer network, wherein each particular resource group, of the plurality of resource groups, represents at least one of: an attribute, a functionality, or a purpose, wherein the set of one or more partitioning criteria comprises the resource group associated with the particular network entity, wherein the first subset of network entities of the first certificate distribution group is associated with a first resource group of the plurality of resource groups.

6. The media of claim 5, wherein the resource group of each particular network entity is determined based on metadata associated with each particular network entity, wherein the metadata comprises a resource group attribute identifying the resource group corresponding to the particular network entity, wherein for a first network entity of the first subset of network entities, a first metadata element comprises a first resource group attribute, wherein the first resource group attribute identifies the first network entity as corresponding to the first resource group.

7. The media of claim 5, wherein the operations further comprise: selecting a second certificate distribution group, of the plurality of certificate distribution groups, for distribution of the first certificate bundle, wherein the second certificate distribution group comprises a second subset of network entities corresponding to a second fault domain of the plurality of fault domains; subsequent to transmitting the first certificate bundle to the first subset of network entities, transmitting the first certificate bundle to the second subset of network entities.

8. The media of claim 7, wherein the second subset of network entities of the second certificate distribution group corresponds to the first resource group of the plurality of resource groups.

9. The media of claim 7, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first subset of network entities, determining a distribution metric with respect to distribution of the first certificate bundle to the first subset of network entities; determining that the distribution metric meets a distribution criterion; and responsive to the distribution metric meeting the distribution criterion, transmitting the first certificate bundle to the second subset of network entities.

10. The media of claim 9, wherein the distribution metric comprises an error count associated with transmitting the first certificate bundle to the first subset of network entities, the error count indicative of a number or a proportion of network entities from among the first subset of network entities with respect to which an error event associated with the first certificate bundle occurs during a verification period; and wherein the distribution criterion comprises the error count remaining below a threshold during the verification period.

11. The media of claim 9, wherein the distribution metric comprises a distribution count associated with transmitting the first certificate bundle to the first subset of network entities, the distribution count indicative of a number or a proportion of network entities from among the first subset of network entities with respect to which a distribution indicator indicates a successful distribution of the first certificate bundle; and wherein the distribution criterion comprises the distribution count meeting a threshold.

12. The media of claim 5, wherein the operations further comprise: selecting a second certificate distribution group, of the plurality of certificate distribution groups, for distribution of the first certificate bundle, wherein the second certificate distribution group comprises a second subset of network entities corresponding to a second resource group of the plurality of resource groups; subsequent to transmitting the first certificate bundle to the first subset of network entities, transmitting the first certificate bundle to the second subset of network entities.

13. The media of claim 12, wherein the second subset of network entities of the second certificate distribution group corresponds to the first fault domain of the plurality of fault domains.
14. The media of claim 13, wherein the operations further comprise: selecting a third certificate distribution group, of the plurality of certificate distribution groups, for distribution of the first certificate bundle, wherein the third certificate distribution group comprises a third subset of network entities corresponding to a second fault domain of the plurality of fault domains; subsequent to transmitting the first certificate bundle to the second subset of network entities, transmitting the first certificate bundle to the third subset of network entities.
15. The media of claim 14, wherein the third subset of network entities of the third certificate distribution group corresponds to the first resource group of the plurality of resource groups.
16. The media of claim 5, wherein the operations further comprise: subsequent to transmitting the first certificate bundle to the first subset of network entities, transmitting the first certificate bundle to one or more additional certificate distribution groups of the plurality of certificate distribution groups; determining a second subset of one or more network entities, associated with the computer network, that are excluded from the plurality of certificate distribution groups, determining, for each particular network entity of the second subset of one or more network entities, a particular aft-distribution group from among a plurality of aft-distribution groups based on the particular fault domain of the particular network entity, selecting a first aft-distribution group, of the plurality of aft-distribution groups, for distribution of the first certificate bundle, wherein the first aft-distribution group comprises a first subgroup of network entities corresponding to the first fault domain of the plurality of fault domains; transmitting the first certificate bundle to the first subgroup of network entities.
17. The media of claim 1, wherein the operations further comprise: determining, based on one or more pre-distribution criteria, a subgroup of one or more network entities, of the plurality of network entities, corresponding to a pre-distribution group; and prior to transmitting the first certificate bundle to the first subset of network entities: transmitting the first certificate bundle to the subgroup of one or more network entities of the pre-distribution group; and determining a successful distribution of the first certificate bundle with respect to the pre-distribution group.
18. The media of claim 1, wherein the operations further comprise: selecting a second certificate distribution group, of the plurality of certificate distribution groups, for distribution of the first certificate bundle, wherein the second certificate distribution group comprises a second subset of one or more network entities corresponding to a second fault domain of the plurality of fault domains; subsequent to transmitting the first certificate bundle to the first subset of network entities: detecting a second trigger condition to transmit the first certificate bundle to the second subset of one or more network entities, wherein the second trigger condition comprises at least one of: a time interval having elapsed subsequent to transmitting the first certificate bundle to the first subset of network entities, or a release phase for releasing the first certificate bundle to the second subset of one or more network entities having commenced; and responsive to detecting the second trigger condition, transmitting the first certificate bundle to the second subset of one or more network entities.
19. The media of claim 1, wherein the operations further comprise: determining, for each particular network entity associated with at least one of the plurality of certificate distribution groups, a particular certificate distribution subgroup, from among a plurality of certificate distribution subgroups, corresponding to the particular network entity, wherein the particular certificate distribution subgroup is determined based on a network address of the particular network entity, at least by applying a randomization function to the network address of the particular network entity, wherein each particular certificate distribution subgroup comprises a particular subgroup of network entities; selecting a first certificate distribution subgroup, of the plurality of certificate distribution subgroups, for distribution of the first certificate bundle, wherein the first certificate distribution subgroup comprises a first subgroup of network entities; selecting a second certificate

distribution subgroup, of the plurality of certificate distribution subgroups, for distribution of the first certificate bundle, wherein the second certificate distribution subgroup comprises a second subgroup of network entities; wherein transmitting the first certificate bundle to the first subset of network entities comprises: transmitting the first certificate bundle to the first subgroup of network entities; and subsequent to transmitting the first certificate bundle to the first subgroup of network entities, transmitting the first certificate bundle to the second subgroup of network entities.

20. The media of claim 1, wherein the computer network comprise a plurality of availability domains respectively representing a physically separate portion of the computer network with respect to a set of one or more failure modes, wherein each of the plurality of network entities is associated with a particular set of one or more failure modes corresponding to a particular availability domain of the plurality of availability domains: wherein the operations further comprise: determining, for each particular network entity of the plurality of network entities, the particular availability domain associated with the particular network entity, and a particular certificate distribution class, from among a plurality of certificate distribution classes, corresponding to the particular availability domain of the particular network entity; selecting a first certificate distribution class, of the plurality of certificate distribution classes, for distribution of the first certificate bundle, wherein the first certificate distribution class comprises a first set of network entities corresponding to a first availability domain of the plurality of availability domains, wherein the first set of network entities comprise the first subset of network entities; selecting a second certificate distribution class, of the plurality of certificate distribution classes, for distribution of the first certificate bundle, wherein the second certificate distribution class comprises a second set of network entities corresponding to a second availability domain of the plurality of availability domains, transmitting the first certificate bundle to the first set of network entities; subsequent to transmitting the first certificate bundle to the first set of network entities, transmitting the first certificate bundle to the second set of network entities.

21. The media of claim 1, wherein partitioning the plurality of network entities into of the plurality of certificate distribution groups comprises: receiving, from a first network entity associated with the computer network, a first request for certificate bundle distribution; determining, based at least in part on a distribution schedule for distributing the first certificate bundle a first release phase for releasing the first certificate bundle for distribution to the first certificate distribution group, wherein the first release phase has commenced prior to having received the first request; wherein selecting the first certificate distribution group comprises: selecting the first certificate bundle for distribution to the first network entity based at least in part on the first release phase having commenced prior to receiving the first request; and wherein transmitting the first certificate bundle to the first subset of network entities comprises: transmitting the first certificate bundle to the first network entity.

22. The media of claim 1, wherein the computer network comprises a virtual cloud network.

23. A method, comprising: detecting a trigger condition to distribute a first certificate bundle comprising a first set of one or more certificate authority certificates; determining, for each particular network entity of a plurality of network entities associated with a computer network, a fault domain associated with the particular network entity, from among a plurality of fault domains associated with the computer network, wherein each particular fault domain, of the plurality of fault domains, represents at least one single point of failure; partitioning the plurality of network entities into a plurality of certificate distribution groups based on a set of one or more partitioning criteria, wherein the set of one or more partitioning criteria comprises the fault domain associated with the particular network entity, and wherein each particular certificate distribution group, of the plurality of certificate distribution groups, comprises a particular subset of network entities, of the plurality of network entities, wherein the particular subset of network entities is associated with a particular single point of failure corresponding to a particular fault domain of the plurality of fault domains; selecting a first certificate distribution group, of the plurality of certificate distribution

groups, for distribution of the first certificate bundle, wherein the first certificate distribution group comprises a first subset of network entities that is associated with a first fault domain of the plurality of fault domains; transmitting the first certificate bundle to the first subset of network entities; wherein the method is performed by at least one device including a hardware processor.

24. A system, comprising: at least one hardware processor; the system being configured to execute operations, using the at least one hardware processor, the operations comprising: detecting a trigger condition to distribute a first certificate bundle comprising a first set of one or more certificate authority certificates; determining, for each particular network entity of a plurality of network entities associated with a computer network, a fault domain associated with the particular network entity, from among a plurality of fault domains associated with the computer network, wherein each particular fault domain, of the plurality of fault domains, represents at least one single point of failure; partitioning the plurality of network entities into a plurality of certificate distribution groups based on a set of one or more partitioning criteria, wherein the set of one or more partitioning criteria comprises the fault domain associated with the particular network entity, and wherein each particular certificate distribution group, of the plurality of certificate distribution groups, comprises a particular subset of network entities, of the plurality of network entities, wherein the particular subset of network entities is associated with a particular single point of failure corresponding to a particular fault domain of the plurality of fault domains; selecting a first certificate distribution group, of the plurality of certificate distribution groups, for distribution of the first certificate bundle, wherein the first certificate distribution group comprises a first subset of network entities that is associated with a first fault domain of the plurality of fault domains; transmitting the first certificate bundle to the first subset of network entities.
